Sensors & Actuators: A. Physical 375 (2024) 115542

Contents lists available at ScienceDirect

Sensors and Actuators: A. Physical

journal homepage: www.journals.elsevier.com/sensors-and-actuators-a-physical

Frequency and damping analysis of hexagonal microcantilever beams

Sai Kishore Jujjuvarapu [a], Lalsingh Devsoth [a], Ashok Akarapu [b], Prem Pal [c], Ashok Kumar Pandey [a,*]

[a] Department of Mechanical and Aerospace Engineering, Indian Institute of Technology Hyderabad, Kandi, Sangareddy, 502284, Telangana, India
[b] Department of Physics, School of Sciences, Malla Reddy University, Hyderabad, 500100, Telangana, India
[c] Department of Physics, Indian Institute of Technology Hyderabad, Kandi, Sangareddy, 502284, Telangana, India

ARTICLE INFO

Keywords:
Hexagonal beam
Natural frequency
Quality factor
MEMS
Microbeam
MEMS accelerometer
Boundary element method

ABSTRACT

Micro-electromechanical systems (MEMS) devices contain various mechanical components to have mechanical movement of the MEMS structure, such as the microcantilever beam. In this work, we present the new design of a microcantilever beam based on the regular hexagon shape and the conventional microcantilever beam. Experimental, analytical, and numerical analyses have been carried out for both configurations. Silicon dioxide (SiO_2) based microcantilever beams are fabricated using wet bulk micromachining. The thickness and width of the beams are kept constant at 0.96 μm and 35 μm, respectively. In contrast, the length of the beam varies based on the number of hexagonal unit cells used. Here, we considered single, double, triple, quadruple, and quintuple hexagonal microbeams for the analyses. The new designs are analyzed for natural frequency, maximum deflection, and Q-factor in the first transverse vibration mode and compared with the equivalent uniform rectangular beams. Due to reduced stiffness, the first mode natural frequency of hexagon beams is lower than the equivalent uniform beams. The static deflection analysis has been performed numerically by applying a normal load of 10 nN at a free end of the beams. The maximum deflection of quadruple and quintuple hexagonal beams is found 450% and 438%, respectively, higher than the uniform beams due to a reduction in stiffness. In addition, the quality factor has been obtained for various damping mechanisms in ambient conditions. From the results obtained, it is observed that the quality factor associated with drag force damping is more dominant over other damping mechanisms. The experimentally measured quality factor has been compared with semi-analytical boundary element methods (BEM) and finite element based method in ANSYS. The values were found to be of the same order of magnitude. Due to the higher flexibility and excellent damping characteristics, the proposed hexagonal microbeams have potential applications as MEMS devices. However, its length and number of arrays can be optimized to minimize the sagging of the beam to the bottom substrate at the end. After analyzing the influence of pressure on structures consisting of hexagon beams and square membrane experimentally and numerically, we investigated the potential application of the suggested hexagonal beams as suspension springs in in-plane as well as out-of-plane MEMS accelerometers across different widths of the microbeams.

1. Introduction

Micro-electromechanical systems (MEMS) are devices that integrate both mechanical and electrical elements. Demand for MEMS devices has increased tremendously in recent years due to advancements in fabrication, miniaturization, and low power requirement [1]. MEMS devices are used as sensors as well as actuators [2]. Recently, micro cantilever-based MEMS sensors have received greater attention in the field of biological [3–9], biomedical [10–16], gas sensing or chemical detection [17–25], and mass sensing [26–29]. One of the critical components in the design of MEMS devices is the mechanical structure. Most MEMS contain mechanical structures as movable components; one such key component is the microcantilever structure. In recent years, many works have focused on the design and analysis of microcantilever structures due to their simple geometry, ease of fabrication, higher sensitivity, and immense applications in various fields. From the mechanical design point of view, the resonance frequency and quality factor are the two important parameters for finding the sensor performance and sensitivity. However, the choice of geometry and shape greatly influences the natural frequency and quality factor of the sensor. The ability of MEMS devices to change shape or respond to external stimuli is a fundamental aspect of their functionality. These shape changes can be harnessed for a wide range of applications.

Metamaterials are often referred to as a class of engineered materials designed to have properties that are not found in conventional

* Corresponding author.
E-mail addresses: me18resch11003@iith.ac.in (S.K. Jujjuvarapu), ashok@iith.ac.in (A.K. Pandey).

https://doi.org/10.1016/j.sna.2024.115542
Received 31 January 2024; Received in revised form 11 May 2024; Accepted 30 May 2024
Available online 4 June 2024

materials [30]. They are artificially structured composites created by arranging multiple elements or constituents at a smaller scale than the wavelength of the phenomena they are intended to manipulate. This unique structure gives them the ability to control and manipulate electromagnetic waves, sound waves, or other physical properties in ways that are not possible with naturally occurring materials [31]. A novel metamaterial created through 4D printing of shape memory polymers, featuring zero Poisson's ratio and intelligent switching mechanical properties in response to temperature changes. This metamaterial also demonstrates effective vibration isolation, reducing amplitudes in specific frequency bands, offering a new approach for low-frequency vibration isolation compared to traditional linear springs [32]. The ability to engineer material properties at the subwavelength scale opens up opportunities for innovation in various fields, with ongoing research and development continuously expanding their potential applications [33]. Shimada et al. [34] have reported the concept of ferroelectric nano-metamaterials and demonstrated its potential through experimental simulations, showcasing complex domain patterns (square, honeycomb, triangular, etc.) achievable through hierarchical nanostructures, thereby opening new avenues for multifunctional device design and advancing metamaterials research. Chen et al. [35] have introduced hierarchically architected metamaterials, replacing regular honeycomb cell walls with hexagonal, kagome, and triangular lattices, demonstrating improved heat resistance and thermal anisotropy, controllable by geometric parameters, alongside enhanced mechanical properties. Usta et al. [36] studied the hierarchical honeycombs with novel slotted and asymmetrical edge cellular shapes, including auxetic and non-auxetic configurations, fabricated using 3D printing with PLA plastic and tested under edgewise compression, evaluating their mechanical properties through experimental and numerical analyses, revealing enhanced strength, stiffness, energy absorption, and stability during deformation. Xu et al. [37] have utilized the novel hierarchical auxetic-hexagonal honeycombs (AuxHex) with various substructures (regular, triangular, and double arrowhead), establishing a mechanism-based method for analyzing plastic collapse stress, which, demonstrated through in-plane compression tests and finite element simulations, enhances load-bearing capacities and energy absorption abilities compared to regular structures, offering insights into optimizing mechanical properties by adjusting secondary structures. Additionally, shape optimization has found various applications, such as acoustic energy absorption [38–42] and soft robotics [43]. It is found that different shapes are widely used for designing metamaterials arranged in periodic form to create 3-dimensional geometry at the macroscale. Therefore, it is important to find appropriate shapes for the cantilever at the microscale in order to achieve the desired characteristics.

For the last two decades, there has been some focus on varying the shape and geometry of microcantilever beams for achieving the higher performance of the cantilever-based MEMS sensors [44–47]. One of the widely explored shapes is the hexagon in which the microcantilever beams resemble the structure of two-dimensional graphene with interconnected hexagons as the building blocks of the microbeams like perforated microbeams [44]. Hemed et al. [48] developed a closed-form analytical model to find the buckling and static deflection of perforated microbeams. To identify the quality factor associated with squeeze film damping, Pasquale et al. [49] performed experiments on silicon and gold perforated plates. Bourouina et al. [50] investigated the influence of thermal loads on the nonlocal resonance frequency of simply supported perforated nanobeams. A closed-form expression of squeeze film damping for perforated microplates under torsional oscillations has been developed by Li et al. [51,52]. Ansari et al. [53] have performed the numerical analysis for vibrational characteristics and stress of the cantilever beam with rectangular, triangular, and stepped profiles. They further stated that the triangular and stepped cantilever beams have better sensitivity than the beam with rectangular profile. Liu et al. [54] used frequency shifts caused by absorbed gases on different geometries of the microcantilever as the measurable quantity for the gas sensor. Parsediya et al. [55] presented a stepped microcantilever beam for detecting the submicron level of glucose and triglyceride, respectively. They also pointed out that the stepped microcantilever beams provide better performance and sensitivity than the conventional rectangular beams for the same surface area, thickness, and length due to the low value of pull-in voltage and less power requirement. Hosseini et al. [56] have developed an analytical model to calculate the natural frequencies of a bimorph v-shaped cantilever beam for a piezo energy harvester. Singh et al. [57] have explored the potential application of a non-uniform microcantilever with linear and quartic variation in beam width as the biosensor for mass sensing applications. They stated that the mass sensitivity of the microcantilever beam is increased by changing the geometry of the beam as compared to the uniform beam. Singh et al. [58] have done theoretical pull-in and modal analysis on the non-uniform microcantilever for highly sensitive MEMS sensors and actuators. Recently, Ashok et al. [59] have designed a microcantilever beam with an arrow shape and studied the effect of rectangular step length and free end width of the beam on the natural frequency and static deflection of the beam. Furthermore, they analyzed the effect of bottom gap and rectangular step length on the quality factor. In extension to the previous work, micro-cantilever beams with stepped trapezoidal geometry have been analyzed for natural frequency and quality factors [60]. Devsoth and Pandey [61] have recently developed an analytical boundary element method (BEM) and ANSYS based numerical technique to compute the hydrodynamic forces of a non-uniform microcantilever beam with linear and quartic variation in width oscillating in incompressible fluid. They stated that a quartic converging beam gives a higher quality factor than a conventional rectangular microcantilever beam. Recently, Gangele et al. [44] have performed a frequency analysis of hexagonal perforated microbeams covered with nanofiber mat. It is found that the hexagonal perforated microbeams provide higher flexibility compared to uniform beams without perforations. However, the focus was limited to investing the frequency tuning effect in detail for only hexagonal structures. Recently Wang et al. [62] utilized the hexagonal-shaped based mechanical metamaterials to design a cantilever probe microstructure for an optical fiber micro force sensor, demonstrating an adjustable elastic constant and ultra-high force sensitivity, with potential applications in biosensing and special scanning tunneling microscope probes. This shows the significance of hexagonal geometry as the functional unit in optimizing mechanical properties and advancing multifunctional device design across various fields.

The above literature shows that the changes in shape and geometry of microcantilever beams significantly influence the sensitivity of cantilever-based sensors. However, there is a lack of systematic study connecting the shape with important dynamic characteristics. Hence, recognizing the importance of shape changes, we first describe the evolution of rectangular-shaped beams into four different shapes of microcantilever beams. These configurations include rectangular, regular hexagonal, irregular hexagons with different perforation geometries and arrow shapes with triangular holes. Subsequently, we present a detailed study to find the influence of the shape on the frequency and different damping phenomena of the beams. Here, the damping mechanisms include drag, squeeze, anchor loss, and thermoelastic damping (TED). The analysis corresponds to the first transverse bending vibration mode of different beam configurations performed. To expand our study, we performed experimental characterizations and computational analyses, specifically examining the transverse bending mode frequency and damping characteristics across different geometries.

To do the analysis, we first discuss the design of various microbeams with simple kinematics movements. After describing the mathematical model to find the first transverse mode natural frequency of uniform cantilever beam, the various dissipation mechanisms are described to find the corresponding quality factor using simple analytical as well

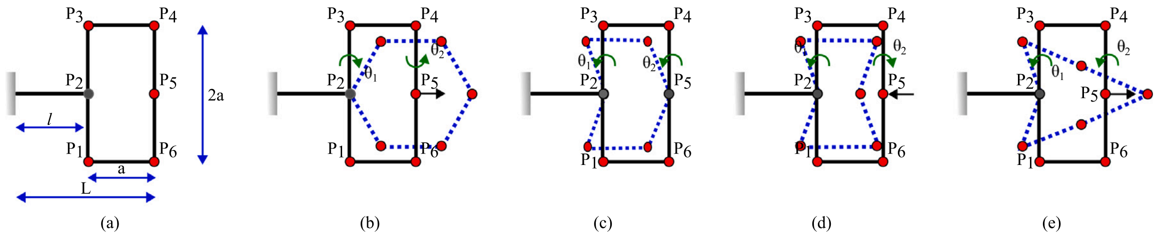

Fig. 1. Schematic representation of (a) rectangular, (b) hexagonal, (c) irregular hexagon-1, (d) irregular hexagon-2 and (d) arrow shape mechanism (Note: Schematics are not drawn on the scale.).

as numerical models. A detailed experimental procedure is described to fabricate the various microcantilever beams using silicon wet bulk micromachining. After presenting the experimental modal analysis, the results and discussion are presented for different shaped beams and structures consisting of beams and membrane at the center. Finally, we explored the potential application of hexagonal microbeams in MEMS accelerometers.

2. Design and modeling of microbeam

In this section, we discuss the evolution of different geometric shapes from the basic rectangular shape using simple kinematic movements of links. This helps us to understand how simple movements can create different and more complex geometric forms starting from a basic rectangle shape.

Consider a rectangular-shaped beam illustrated in Fig. 1(a) as a kinematic mechanism comprised of distinct links interconnected at nodes (P_1 to P_6). These nodes are depicted as free revolute joints, permitting rotational movement of the links. Each link is denoted by the length a, and the overall length of the beam is denoted as L. An additional beam of length l is affixed to the rectangular section at node P_2, with its other end connected to a fixed support. Through an outward sliding movement at node P_5, links P_2P_3 and P_4P_5 undergo opposite rotational motions expressed as θ_1 (clockwise (CW)) and θ_2 (anticlockwise (ACW)),respectively while keeping the nodes P_3, P_4 and P_1, P_6 are on the same line. This results in the formation of a hexagonal shape, as depicted by the dotted lines in Fig. 1(b). Further, by fixing the nodes P_2 and P_5 at their initial location and giving a rotational movement to the links P_2P_3 and P_4P_5 in the same ACW direction represented by θ_1 and θ_2. This results in a hexagonal geometry of different shapes called irregular hexagon-1 (IRHB-1), which is shown in Fig. 1(c) as in dotted lines. Subsequently, by having an inward sliding movement at node P_5 and keeping node P_2 as fixed at its initial location, the links P_2P_3 and P_4P_5 undergo rotations in opposite directions denoted as θ_1 in ACW and θ_2 in CW direction, which form a shape of hexagon called irregular hexagon-2 (IRHB-2) as shown in Fig. 1(d). At last, similar to Case 1 (hexagon), when the node P_2 is fixed in its initial position, and rotational movements are imparted to the links P_2P_3 and P_4P_5 in the same anticlockwise (ACW) direction, coupled with an outward sliding motion at node P_5, an arrow-shaped beam is generated. This configuration is depicted in Fig. 1(e) through dotted lines. This design of the dynamic kinematic system reveals its remarkable flexibility and transformative nature, paving the way for various geometric configurations. This has a significant application in MEMS technology and microbeam design. To compare the results of hexagonal beams with those of uniform beams, we present the mathematical models of uniform beams to obtain frequencies and quality factors.

2.1. Modal analysis

Consider an equivalent uniform rectangular microcantilever beam as shown in Fig. 2(b), with length (L), width (w), and thickness (t). To find the first transverse mode natural frequency of the microcantilever beam, the exact mode shape $\psi(x)$ is assumed as follows,

$$\psi(x) = Acos(\lambda x) + Bsin(\lambda x) + Ccosh(\lambda x) + Dsinh(\lambda x). \tag{1}$$

Here, A, B, C, and D are constants that can be found from the boundary conditions. For a cantilever beam fixed at one end and free at the other end, the boundary conditions are defined as,

$$\psi(0) = 0, \psi'(0) =, \psi''(L) = 0, \psi'''(L) = 0. \tag{2}$$

From the boundary conditions and assumed mode shape, the total effective mass (m_{eff}) and total effective stiffness (k_{eff}) of the beam can be found as,

$$m_{\text{eff}} = \int_0^L m\psi(x)^2 dx \tag{3}$$

$$k_{\text{eff}} = \int_0^L \psi(x)\frac{d^2}{dx^2}\left(EI\frac{d^2\psi(x)}{dx^2}\right)dx \tag{4}$$

Hence, the first transverse bending mode frequency is found as,

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{\text{eff}}}{m_{\text{eff}}}} \tag{5}$$

From the Eqs. (3) and (4), the m_{eff} and k_{eff} of a microcantilever beam are calculated as,

$$m_{\text{eff}} = 0.26\ \text{m};\ K_{\text{eff}} = \frac{3EI}{L^3} \tag{6}$$

2.2. Quality factor analysis

Many MEMS oscillating micro beams/systems interact with the surrounding medium or system through energy transfer from one form to the other. The net useful energy of the system is found from the algebraic sum of the input energy required and the amount of energy dissipated from the system. MEMS structures are subjected to different energy dissipation mechanisms due to various damping effects in the system, such as Stoke's flow (drag force damping), squeeze-film damping, anchor loss, and thermoelastic damping. The quantity known as the quality factor (Q) is used to calculate the impact of energy dissipation on the dynamic system's frequency response. For a dynamic system with input energy (E_{input}), vibrating with the natural frequency (ω_n), and the influence of energy dissipation per cycle can be quantified by the quality factor as [63],

$$Q = 2\pi\frac{E_{\text{input}}}{\Delta E}, \tag{7}$$

here, E_{input} is the energy input per cycle, and ΔE is the energy dissipation per cycle. Hence, from Eq. (7), it is evident that the quality factor of the dynamic system is inversely proportional to the amount of energy dissipated from the system. In another way, the quality factor measures the sharpness of the system's frequency response curve, which reduces as the energy losses in the system increase. Hence, by considering all

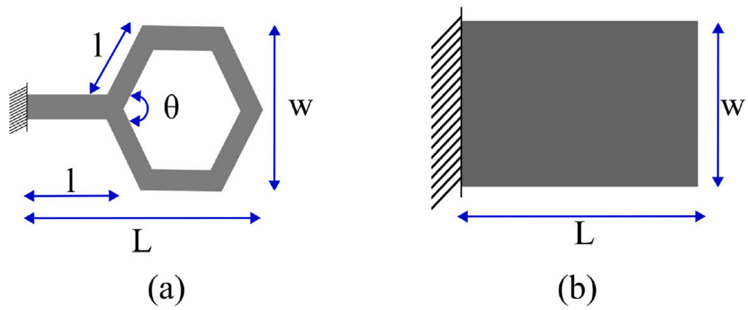

Fig. 2. Schematic of (a) single hexagonal shaped microcantilever beam with θ as the angle of hexagon (120°) angle and side length as l (b) Equivalent uniform rectangular beam with length L and width w.

forms of energy interaction of the system, the overall quality factor is defined as,

$$\frac{1}{Q_{\text{Total}}} = \frac{1}{Q_{\text{AL}}} + \frac{1}{Q_{\text{TED}}} + \frac{1}{Q_{\text{Fluid}}} + \frac{1}{Q_{\text{Other}}}, \tag{8}$$

here, Q_{AL}, Q_{TED}, Q_{Fluid}, and Q_{Other} are the quality factor corresponding to anchor loss, thermoelastic damping (TED), drag and squeeze-film damping, and other damping mechanisms like surface loss, respectively. We knew that the influence of different damping mechanisms would affect the system's dynamic response differently and that the associated quality factor would have varying orders of magnitude. Hence, Eq. (8) is rewritten as,

$$\frac{1}{Q_{\text{Total}}} = \sum_j \frac{1}{Q_j} \approx \frac{1}{Q_{\text{dominant}}}, \tag{9}$$

here, Q_{dominant} is the most dominating value of the quality factor corresponding to the most dominating damping mechanism through which the energy loss is higher under the given operating conditions of the dynamic system. Further, the microcantilever beam's total quality factor can be determined by considering various damping mechanisms, as shown below.

- **Anchor Loss**: Anchor loss is also called clamping or support loss. It involves both the substrate and the oscillating microbeam; whenever the beam oscillates, part of the vibrational energy of the beam in the form of elastic waves will propagate into the infinite substrate medium. The energy dissipated into the substrate is irreversible and considered lost. Hence, the quality factor (Q_{AL}) corresponds to the anchor loss for a cantilever microbeam as shown is given by Judge et al. [64] as,

$$\frac{1}{Q_{\text{AL}}} = A\frac{w}{L}\left(\frac{t_s}{L}\right)^4, \tag{10}$$

  where the basic assumptions involved in using the above equation are as follows,

  - The thickness of the substrate (h_s) is larger than the thickness of the microbeam (t_s).
  - The substrate is considered as an infinite medium to absorb the elastic waves coming from the microbeam through fixed support, i.e., no reflection of elastic waves from the substrate and considered as lost.

- **Stoke's flow damping**: Stoke's flow damping is also called drag force damping or loss due to airflow. The airflow around the microbeam functions as a damping medium whenever the microbeam is operated under ambient pressure (101.325 kPa) conditions by exerting drag force on the surface of the beam, which oscillates in isolation. The influence of the drag force damping is significant when the air gap between the static boundaries and structure is higher than the width of the microbeam. Considering a microbeam oscillating with the natural frequency of ω_n in ambient environmental conditions, the quality factor associated with loss due to drag force damping is given as [63,65],

$$Q_{\text{drag}} = \frac{\rho_s w t_s \omega_n}{8\mu}, \tag{11}$$

  here, ρ_s, w, t_s, and μ are the density, width, thickness of the microbeam, and dynamic viscosity of the air, respectively.

- **Squeeze-film damping**: It involves both the microbeam and a fixed substrate, and it appears whenever the microbeam oscillates in a direction normal to the fixed substrate by squeezing out the thin film of air between them. Squeeze-film damping greatly influences the dynamic behavior of the microbeam whenever it oscillates very near the fixed substrate. For a microbeam, the quality factor corresponding to the squeeze film damping is given as [66],

$$Q_{\text{squeeze}} = \frac{\rho_s g^3 t_s \omega_n}{\mu w^2}, \tag{12}$$

  where g is the air gap thickness between the microbeam and fixed substrate. The squeeze film damping will be more dominant than the drag force damping whenever the air gap thickness is smaller than one-third of the width of the beam.

- **Thermoelastic damping (TED)**: It is a bulk loss in an oscillating beam due to the change in temperature within the beam as a result of the elastic strain experienced. The temperature gradient causes the conversion of elastic strain energy into irreversible heat energy, which is conducted through the body. The heat transfer due to conduction results in energy loss in the oscillating microstructure. This phenomenon dampens the motion of the beam. The quality factor (Q_{TED}) for a microcantilever beam associated with the thermoelastic damping found as [67],

$$\frac{1}{Q_{\text{TED}}} = \frac{E_s \alpha T_0}{C_p \rho_s}\left(\frac{6}{\zeta^2} - \frac{6}{\zeta^3}\cdot\frac{\sinh(\zeta)+\sin(\zeta)}{\cosh(\zeta)+\cos(\zeta)}\right), \tag{13}$$

  here, E_s, α, T_0, and C_p are the elastic modulus, thermal expansion coefficient, ambient temperature, and specific heat of the beam material, respectively. Also, the term ζ is given by,

$$\zeta = w\sqrt{\frac{C_p \rho_s \omega}{2\kappa}}, \tag{14}$$

where κ and ω are the thermal conductivity and angular natural frequency of the microbeam. Thermoelastic damping can be both beneficial and detrimental, depending on the application. In some cases, it can be used to enhance damping in MEMS resonators, improving their stability and reducing unwanted vibration modes. However, thermoelastic damping might need to be mitigated or minimized in precision applications where minimal damping is desired. Hence, we need to consider thermoelastic damping effects, especially when designing resonators or other components sensitive to mechanical vibrations. Another type of damping mechanism, known as surface loss damping, is attributed to the development of a thin oxide layer on the microbeams surface. This oxide layer induces an increase in surface roughness, contributing to energy losses in the microbeam system. When the microbeam undergoes movement or vibration, the resulting friction and surface effects lead to energy dissipation. For this research, microbeams with a thickness of 1 μm were created using silicon dioxide (SiO_2) as the structural material through a process of wet bulk micromachining. Since the SiO_2 microbeams were fabricated as a monolayer on the silicon substrate, we excluded the consideration of surface loss damping in our analyses, recognizing its minimal contribution to energy dissipation [68,69]. Nevertheless, future investigations will explore a more detailed exploration of the impact of surface loss damping on the dynamic characteristics of the microbeams.

2.3. Numerical modeling

In this section, we elaborate on the numerical modeling approach we employed to determine the natural frequency and quality factor of hexagonal beams as well as their equivalent uniform rectangular counterparts. For the present study, we considered silicon dioxide (SiO_2) as the structural material with material properties, Young's modulus (E) is 66.26 Gpa, Poisson's ratio is 0.3, and density is 2200 kg/m^3. We utilized CoventorWare software based on the finite element method to accomplish this. The natural frequency and associated quality factor evaluation involve the MemMech solver, which calculates thermoelastic damping and anchor loss damping. Additionally, the DampingMM solver derives the damping coefficient (C_d) concerning drag force damping and squeeze film damping. Our methodology employs extruded bricks with parabolic elements, featuring three nodes along each mesh edge, a central node for every mesh face, and an extra node at the center of the element. This specific arrangement results in a 27-node hexahedral structure, enhancing accuracy by providing each node with 6 degrees of freedom. For simulations involving anchor loss, we established quiet boundary conditions towards the substrate, thereby allowing infinite space for absorbing elastic wave energy transmitted through supports or anchors. In the present investigation, we calculated the squeeze film damping coefficient for all hexagonal and equivalent uniform micro beams, accounting for a 80 μm substrate gap. However, one must be careful while choosing the meshing parameter for DampingMM to compute the drag coefficient. Our study found that hexahedral with liner elements using split and merge algorithms provides accurate results.

Furthermore, we extended our approach to incorporate the boundary element method (BEM) for uniform beams. We also applied a numerical technique described in a previous study to compute the quality factor related to drag force damping as detailed in Ref. [61].

3. Experimental procedure

In the present work, we used a single-side polished, 4-inch diameter, p-type, (100) oriented silicon wafer. The thermally grown oxide film is used as a structural and masking layer when the microcantilever beams are fabricated using a wet bulk micromachining process in wet anisotropic etching. Fig. 3 illustrates the process steps used in the fabrication of silicon oxide (SiO_2) microcantilever beams. Firstly, SiO_2 layer of about 1 μm thick is deposited using the thermal oxidation technique. Thereafter, the oxidized silicon wafer is patterned using UV photolithography with the help of a Mask Aligner (Midas MDA 400M). The oxide layer is selectively etched in buffered hydrofluoric acid (BHF) solution where the positive photoresist is used as the mask layer. Now, the wafer is diced into small chips of size 2 × 2 cm^2. Subsequently, the diced chips are sequentially cleaned in acetone and piranha solution (H_2SO_4:H_2O_2:: 1:1) for the removal of photoresist and organic contaminants. Each cleaning step is followed by rinsing in deionized (DI) water. A very thin chemical oxide is grown in a piranha bath, which delays the silicon etching in an alkaline solution. Hence, the samples are dipped in 1% hydrofluoric acid (HF) solution for 30 s to remove the chemical oxide. This step is followed by thorough rinsing in DI water. Finally, the microstructures are fabricated using wet bulk micromachining in 25 wt% Tetramethylammonium hydroxide (TMAH) (99.999%, Alfa Aesar) at 74.5 ± 0.5 °C. After the completion of the etching process, the chips are rinsed in DI water and then dried using an air blower. The optical micrograph of the fabricated uniform microcantilever beam is shown in Fig. 3(k). After the fabrication of oxide cantilever beams, the thickness of the oxide layer is measured using ellipsometry. It was measured to be 0.965 μm.

From the above micro fabrication procedure, we have developed single, double, triple, quadruple, and quintuple hexagonal microcantilever beams and their equivalent uniform rectangular microcantilever beams as shown in Fig. 4. For the current study, we have considered a single hexagonal microcantilever beam as a building block, as shown in Fig. 2(a). The length (L), thickness (t_s) and width (w) of the single hexagonal unit are fixed as 55 μm 0.965 μm and 35 μm, respectively. The length of a single hexagonal unit is fixed, while the lengths of the proposed microcantilever beams vary depending on the number of hexagonal unit cells used in the beam. Fig. 4 shows the optical images of proposed hexagonal microcantilever beams of length (Fig. 4:(a)–(e)) along with their equivalent uniform rectangular microcantilever beams (Fig. 4(f)–(j)). However, we have found that the quadruple and quintuple beams consisting of four and five hexagonal unit cells are not released due to the sagging of free ends to the bottom substrate as shown in Fig. 4(d) and (e). It is confirmed by observing its bending from the fixed to its free end by varying the position of the lens of the optical microscope downward towards the beam surface, as shown in Fig. 5(b). Finally, the experimental frequency response curves of the suspended beams are measured using the scanning laser vibrometer (Polytech, OFV-5000 vibrometer) [59] as shown in Fig. 5(a). Due to the sharp bending of the beam near the fixed edge of quadruple and quintuple hexagonal beams, the laser becomes out of focus to sense the reflecting light, as shown in Fig. 5(b). Therefore, quadruple and quintuple hexagonal beam frequencies cannot be measured experimentally.

4. Results and discussion

In this section, we analyze the first transverse mode natural frequency, static deflection, and quality factors associated with fluid damping, anchor loss, and thermoelastic damping of hexagonal and equivalent rectangular microcantilever beams.

4.1. Modal analysis

The first transverse mode natural frequency of the regular hexagonal microcantilever beams (HB) and the equivalent uniform rectangular beams (UB) are measured in an ambient (101.325 kPa) environment using a laser scanning vibrometer (Polytec, OFV-5000 vibrometer). The comparison of the first transverse mode natural frequency of single (HB-1), double (HB-2), triple (HB-3), quadruple (HB-4), and quintuple (HB-5) hexagonal microcantilever beams with equivalent uniform rectangular microcantilever beams is shown in Fig. 6(a). From Fig. 6(a), it is clear that the transverse mode natural frequency of the hexagonal micro beams is lower than the equivalent uniform rectangular micro beams. From the theory of vibration, it is well known that the natural frequency of the microcantilever beam is directly proportional to the square root of the effective stiffness (spring constant (K_{eff})) of the beam and inversely proportional to the effective mass (M_{eff}) of the cantilever beam $\left(\omega_n = \sqrt{\frac{K_{\mathrm{eff}}}{M_{\mathrm{eff}}}}\right)$. For the proposed micro beams, the decrease in the natural frequency of the hexagonal microcantilever beams, compared to uniform rectangular beams, is due to the marginal decrease in the stiffness resulting from the shape change of microbeams. Further, it is also noted that the natural frequency of the hexagonal microcantilever beams decreases from single to quintuple structures due to a marginal decrease in the stiffness or an increase in the effective mass of the hexagonal microcantilever beams. In addition, the first transverse

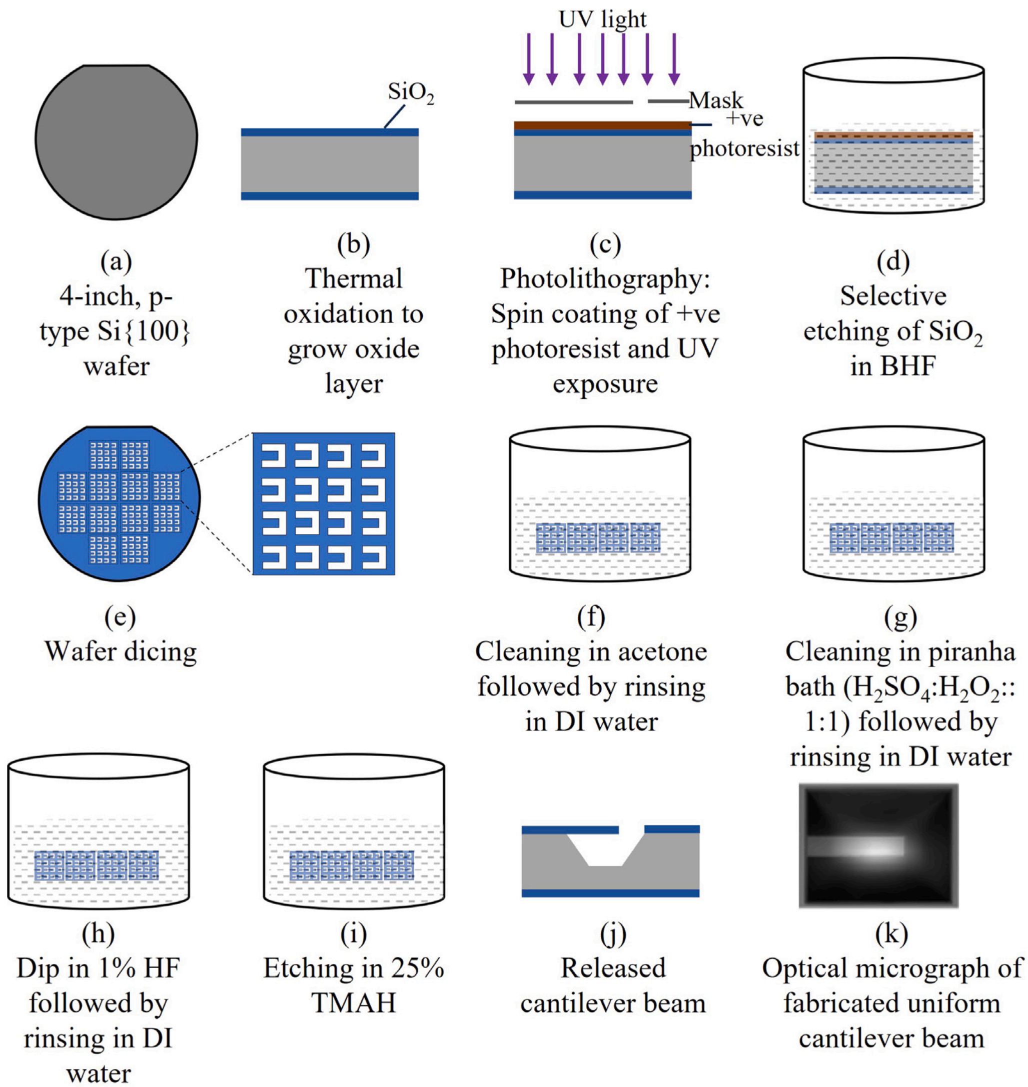

Fig. 3. Schematic views of process steps used to fabricate SiO_2 cantilever beams using silicon wet bulk micromachining.

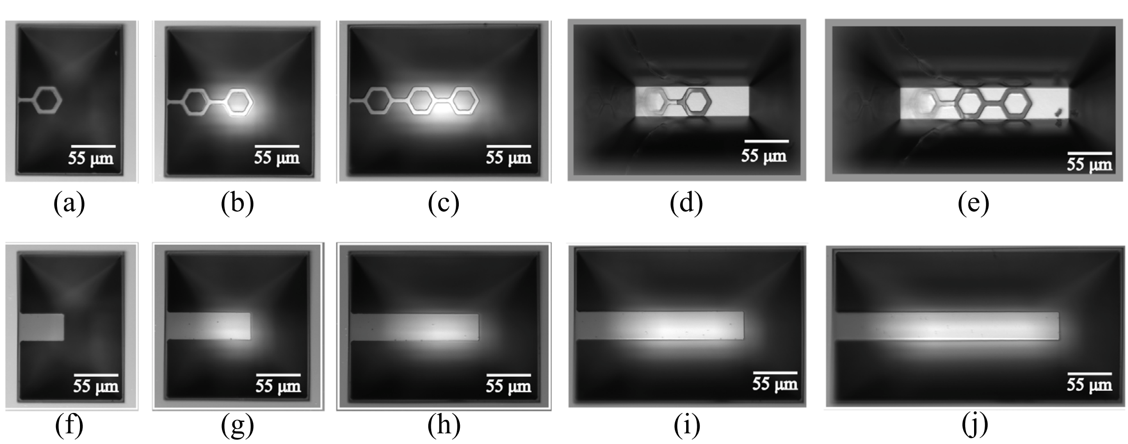

Fig. 4. Optical images of (a) single, (b) double, (c) triple, (d) quadruple, and (e) quintuple hexagonal microcantilever beams with their equivalent uniform rectangular microbeams ((f)–(j)), respectively. Here, note that the images are not up to the actual scale.

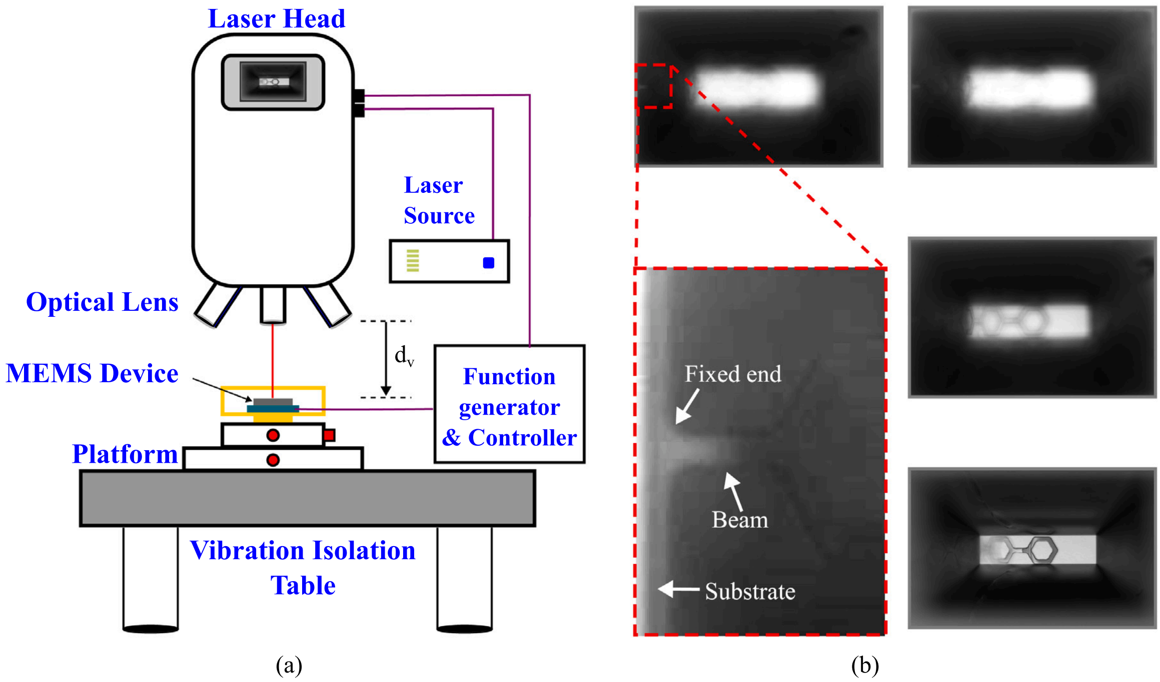

Fig. 5. (a) Schematic of the experimental setup for laser scanning vibrometer (Polytec, OFV-5000 vibrometer showing the focal distance, d_v, between the sample and the lens. (b) Optical images showing the sagging of quadruple (HB-4) microcantilever beams at different positions as it downward towards the sample.

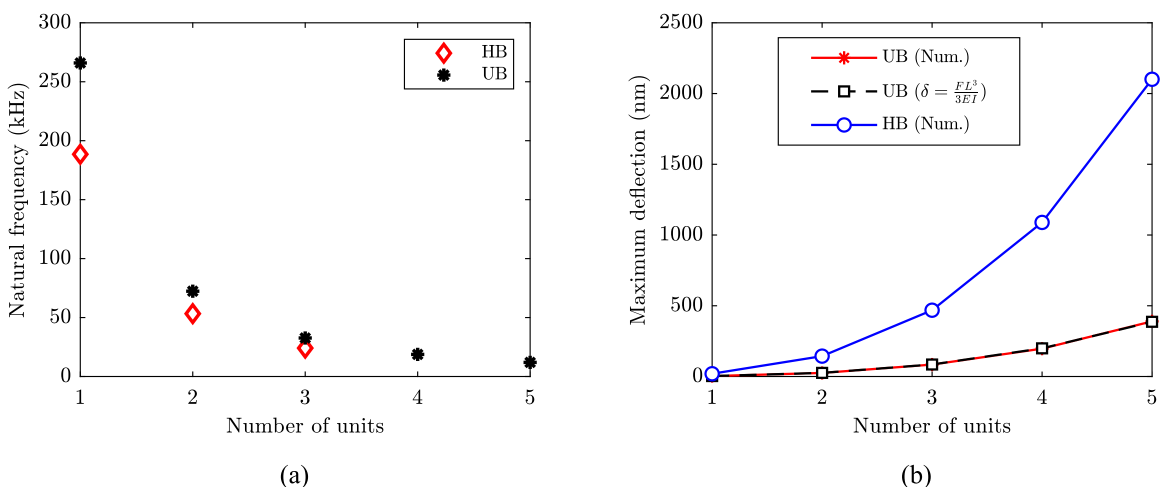

Fig. 6. (a) Comparison of experimental first transverse mode natural frequency of hexagonal microcantilever beams and equivalent uniform rectangular microcantilever beams; (b) comparison of maximum deflection of the hexagonal microcantilever beams and equivalent uniform microcantilever beams for the concentrated load of 10 nN at the free end.

mode natural frequency is computed numerically using CoventorWare software. Both the experimental and numerically computed natural frequency for various beams have been tabulated in Table 1, and it is observed that both numerical and experimental results are shown in good agreement with each other with a maximum percentage error of 10.11%.

4.2. Deflection analysis

The static deflection analysis of the hexagonal and equivalent uniform rectangular beams has been computed numerically using CoverntorWare by applying the concentrated load (F) of 10 nN at the free end of the beams. Fig. 6(b) shows the comparison of maximum deflection experienced by the hexagonal beams and their corresponding equivalent uniform rectangular beams to the applied concentrated load of 10 nN at their free ends. From Fig. 6(b), it is found that the hexagonal

Table 1
Comparison of numerical and experimental transverse mode natural frequency of hexagonal and uniform rectangular microcantilever beams.

Conf.	Frequency (kHz)			
	Numerical	Experimental	Analytical	% Error (Num. & Exp.)
HB-1	186.73	188.52	–	0.94
HB-2	50.01	53.29	–	6.15
HB-3	23.11	24.10	–	4.1
HB-4	13.28	–	–	–
HB-5	8.61	–	–	–
UB-1	292.8	265.9	270	10.11
UB-2	71.97	72.35	67.5	0.52
UB-3	32.11	32.66	30.37	1.68
UB-4	18.23	18.75	17.19	2.77
UB-5	11.58	11.97	11.04	3.26

microcantilever beams have maximum deflection higher than the uniform beams for the same applied load. Furthermore, it is also shown that the maximum deflection of the beams increases as the number of hexagonal units in the beams increases from single to quintuple. Further, we compared the numerical results of deflection of the uniform beam (δ) with the analytical ($\delta = \frac{FL^3}{3EI}$) values, and they are shown good agreement with each other.

From the analysis, the maximum deflection experienced by the quadruple and quintuple hexagon microcantilever beams is 450% and 438%, respectively, higher than the corresponding equivalent uniform rectangular microcantilever beams. The increase in the maximum deflection of the hexagonal microcantilever beams is due to the reduction in the stiffness of the beams as the number of hexagonal units increases. Hence, to achieve better sensitivity in terms of deflection, the hexagonal microcantilever beams are to be considered over the uniform rectangular beams.

4.3. Quality factor analysis

Now, we discuss the influence of various damping mechanisms over the quality factor of the microbeam. For the quality factor analysis, we considered anchor loss (Q_{AL}), fluid damping ($Q_{\mathrm{drag/squeeze}}$), and thermoelastic damping (Q_{TED}) are the main damping mechanisms which influence the dynamic response of the microbeam. Fig. 7 shows the quality factor corresponding to the first transverse mode of the microcantilever beam subjected to the various damping mechanisms. Fig. 7(a) depicts the experimental quality factor measured using a laser vibrometer in an ambient (101.325 kPa) environment. The full width at half maximum (FWHM) method of the frequency response of the beams is employed to measure the value of the quality factor. From the experimental results, it has been found that as the number of units increases, the quality factor decreases. Subsequently, Fig. 7(b) shows the numerically computed quality factor in Coventorware associated with drag and squeeze film damping corresponding to the first transverse mode of the hexagonal and equivalent uniform rectangular cantilever beam. As the hexagon units increase from one to five, the drag and squeeze film damping quality factor (Q_{Drag}) and (Q_{Squeeze}) decreases due to the increase in the surface area. Fig. 7(b) shows that the influence of drag force damping is more dominant over the squeeze film damping for both the hexagonal and equivalent rectangular uniform beams. Further, it is also observed that as the hexagonal beam configuration changes from single to quintuple, the value of the quality factor reduces due to the availability of more surface area for energy dissipation through fluid damping. The Q_{drag}, and Q_{squeeze} are in the order of 10^1–10^2, and 10^3–10^4, respectively. Fig. 7(c) shows the numerically computed quality factor associated with anchor loss corresponding to the first transverse mode of a hexagonal and equivalent uniform rectangular cantilever beam. As the hexagon beam configuration changes from a single hexagon to a quintuple hexagon structure, the anchor loss quality factor (Q_{AL}) increases due to the dissipation of elastic energy through the supports being reduced as the oscillation frequency decreases. The order of (Q_{AL}) is found to be 10^8–10^{12}. In addition, we have also computed the quality factor corresponding to the thermoplastic damping (Q_{TED}), which is shown in Fig. 7(d). The obtained results show that the Q_{TED} increases as the beam configuration changes from single to quintuple because the increase in thermal relaxation time constant corresponds to the oscillation frequency of the beams. The order of Q_{TED} is found as 10^6–10^7.

From the above discussion, we can observe that the influence of drag force damping is more dominant over the other damping mechanisms. Hence, the overall quality factor of the microbeams can be approximated to the Q_{drag} from Eq. (9). Further, we have also validated our experimental quality factor ($Q_{\mathrm{Exp.}}$) with the semi-analytical boundary element method (BEM) and numerical method in ANSYS, proposed by Devsoth et al. [61]. Table 2 shows the comparison of

Table 2
Comparison of semi-analytical, Ansys, Coventorware numerical with experimental quality factor associated with drag force damping (Q_{drag}) of uniform and hexagonal microcantilever beams.

Conf.	Quality factor (Q_{Drag})			
	BEM	ANSYS (Num.)	CoventorWare (Num.)	Exp.
UB-1	88.15	97.12	615	125
UB-2	40.63	43.85	224	85
UB-3	25.22	26.33	121	30
UB-4	17.40	17.27	78	25
UB-5	13.19	12.64	55	20
HB-1	–	51.78	202	70
HB-2	–	18.44	71	27
HB-3	–	9.658	37	18
HB-4	–	5.664	23	–
HB-5	–	3.989	16	–

Q_{drag} found from the BEM, numerical method in ANSYS, and numerical Coventorware with experimental quality factor ($Q_{\mathrm{Exp.}}$). From Table 2, it observed that the Q_{drag} values from BEM, numerical ANSYS, and experimental are shown good agreement with the same order of magnitude; however, Coventorware gives a higher value of quality factor due to the assumptions involved in the usage of DampingMM solver.

Similarly, in Table 3, the 1D analytical model of a uniform rectangular microcantilever beam was used to compare numerical Coventorware results of various quality factors associated with different damping mechanisms. Due to the fundamental assumptions included in the analytical model, the resulting results have shown moderate agreement with numerical results.

4.4. Comparison of different beams

This section discusses the various microcantilever beams with different shapes as the building units, as shown in Fig. 8. Similar to hexagonal microcantilever beams, we have considered arrow and irregular hexagon-shaped beams for further study. The thickness and width of the beams are fixed at 0.965 μm and 40 μm, respectively, while the length varies from ≈60 μm to ≈300 μm. But in the case of arrow-1 (Fig. 8(a)), the length varies from ≈60 μm to ≈199 μm due to the absence of a connecting beam along the length. In the case of irregular hexagon-1 (IRHB-1) and irregular hexagon-2 (IRHB-2), the optical images (Fig. 8(c)–(d)) show that the beams become more flexible as the number of units increases from 3 to 5, while the arrow shape, it is from 4 to 5 units (Fig. 8(a)–(b)). The beams were snapped towards the substrate without any external force due to their flexibility or reduction in stiffness; hence, the higher-length beams were not characterized experimentally. However, numerical computations have been conducted to find the natural frequency and quality factor associated with different damping mechanisms.

To begin, we conducted experiments on various microcantilever beams under ambient conditions (101.325 kPa). Fig. 9(a) presents the experimentally discovered first transverse mode natural frequency of a microcantilever beam with different shapes. The results show that while the number of units or length increases, the natural frequency of different beams decreases due to a reduction in stiffness or an increase in the effective mass of the beams. Further, we also compared the obtained results with hexagons (HB) and their equivalent uniform beams (UB). In the case of shorter beams (1 unit), UB has a higher natural frequency than others due to the higher stiffness. The arrow-1 configuration has a higher natural frequency among the other different shape beams due to the smaller length and higher stiffness.

Similarly, we measure the experimental quality factor using the full width at half maximum (FWHM) method. The experimental quality factor for different microbeams is shown in Fig. 9(b). According to the results, an equivalent uniform beam (UB) has a higher quality

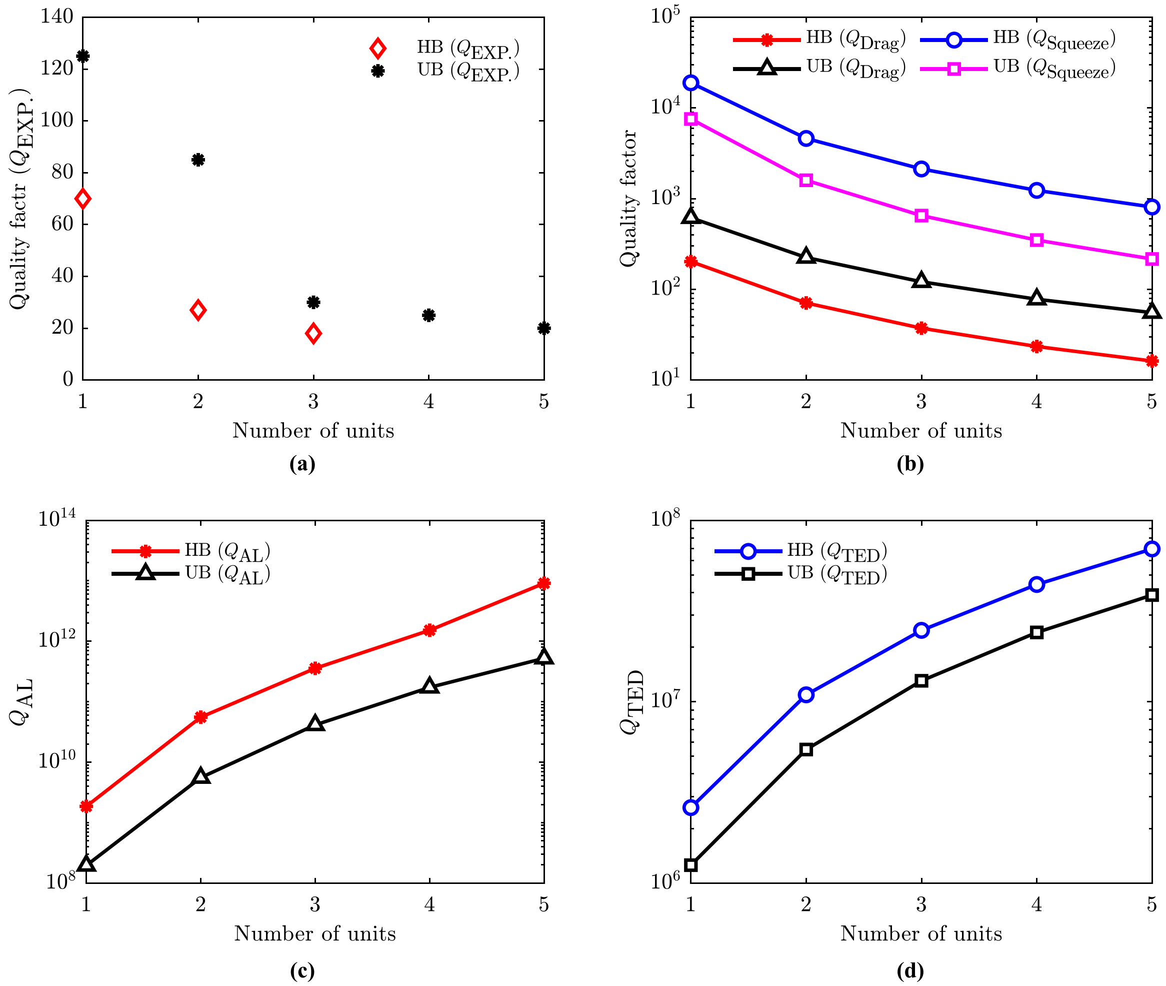

Fig. 7. (a) Comparison of experimental quality factor ($Q_{\mathrm{Exp.}}$) measured in the ambient environment (101.325 kPa), (b) numerically computed quality factor in Coventorware, (Q_{Drag} and Q_{Drag}) corresponds to fluid damping, (c) numerically computed quality factor in Coventorware (Q_{AL}) corresponds to the anchor loss, and (d) numerically computed quality factor (Q_{TED}) in Coventorware corresponds to the thermoelastic damping of hexagonal and equivalent rectangular microcantilever beams.

Table 3
Comparison between numerical and analytical quality factors related to various damping mechanisms of uniform microcantilever beams.

Conf.	Quality factor (Q)							
	Q_{AL}		Q_{Drag}		Q_{Squeeze}		Q_{TED}	
	Num.	Ana. Eq. (10)	Num.	Ana. Eq. (11)	Num.	Ana. Eq. (12)	Num.	Ana. Eq. (13)
UB-1	1.95×10^{9}	5.34×10^{7}	615	918	7556	8.77×10^{5}	1.25×10^{6}	1.47×10^{8}
UB-2	5.55×10^{9}	1.71×10^{9}	224	225	1595	2.15×10^{4}	5.44×10^{6}	3.71×10^{7}
UB-3	4.12×10^{10}	1.26×10^{10}	121	100	650	9.62×10^{3}	1.31×10^{7}	1.69×10^{7}
UB-4	5.22×10^{11}	5.23×10^{10}	78	57	349	5.54×10^{3}	2.41×10^{7}	9.89×10^{6}
UB-5	1.72×10^{11}	1.58×10^{11}	55	36	216	3.47×10^{3}	3.87×10^{7}	6.45×10^{6}

factor than other microbeams. Among the various shapes, arrow-1 has the highest quality factor. We then used Coventorware software to calculate the numerical quality factor associated with various damping mechanisms. Furthermore, as the length increases, the quality factor decreases because the beams have become more flexible; thus, the fluid particles around the beams have enough time to adhere to the surface of the microbeams, causing energy loss through drag force damping. We also computed the numerical quality factor corresponding to the drag force damping, squeeze film damping, anchor loss, and thermoelastic damping using Coventorware, as shown in Fig. 10.

From Fig. 10(a), it has been depicted that as the length increases, the Q_{Drag} of different beams decreases, causing more energy loss through drag force damping at higher lengths. It has also been seen that apart from UB, the arrow-1 configuration has a higher Q_{Drag} among other shapes of microbeams because the arrow-1 has lower flexibility and length than the other beams. Similarly, Fig. 10(b) shows computed the quality factor corresponding to the squeeze film damping (Q_{Squeeze}). For the numerical computation, we have considered a gap of 80 μm between the beam and substrate surface. The results show that Q_{Squeeze} is inversely proportional to the length of the microbeams. Furthermore, at lower beam lengths (1 unit), the hexagonal beam (HB) has a better quality factor than others. This is due to the presence of a hexagonal hole, which causes the fluid to travel through the hole, resulting in decreased energy loss via squeeze film damping. As the number of units increases from 2 to 5, arrows have a higher Q_{Squeeze}

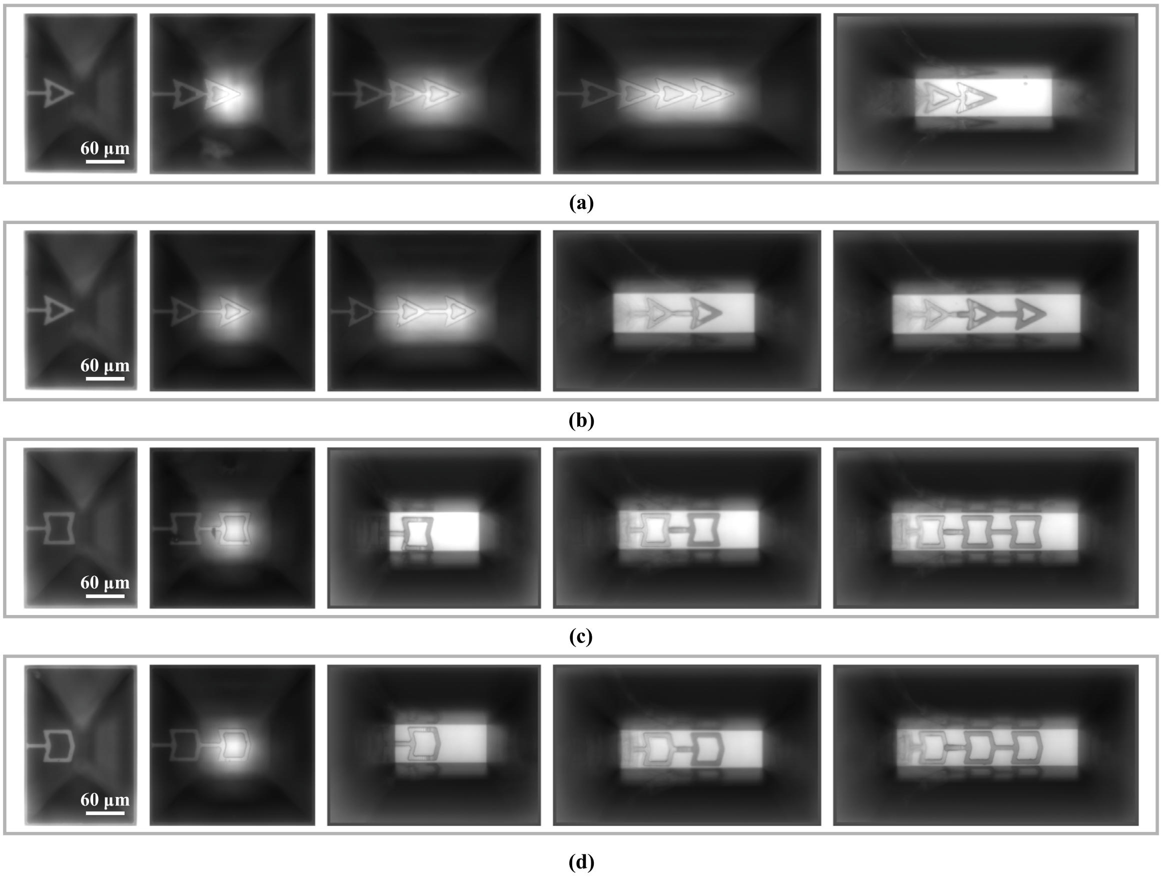

Fig. 8. (a) Optical images of arrow-1, (b) arrow-2, (c) irregular hexagon-1 (IRHB-1), and (d) irregular hexagon-2 (IRHB-2) microbeams, respectively. Here, note that the images are not up to the actual scale.

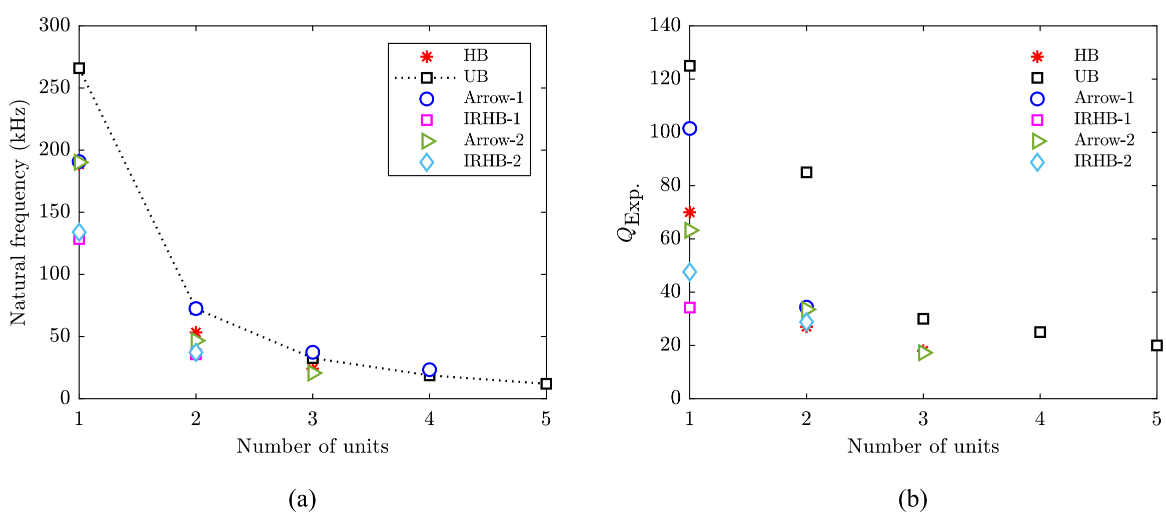

Fig. 9. (a) Comparison of experimental first transverse mode natural frequency; and (b) the experimental quality factor ($Q_{\mathrm{Exp.}}$) of different microcantilever beams.

than other shapes, whereas uniform beams have a lower Q_{Squeeze} because, in the absence of a hole, the fluid gets squeezed between the beam and the substrate surface, causing more energy loss through squeeze film damping than other microbeams.

In addition to the fluid damping, we computed the anchor loss and thermoelastic damping of different microbeams. Fig. 10(c) presents the quality factor corresponding to the anchor loss (Q_{AL}), and it has been noted that the Q_{AL} is directly proportional to the length of microbeams. From Fig. 10(c), it has also been seen that the order of increasing Q_{AL} is as follows: UB < arrow-1 < HB < arrow-2 < IRHB-2 < IRHB-1, this is because Q_{AL} is inversely proportional to the stiffness of the beams. As the beam becomes flexible, the amount of energy carried through elastic waves and getting lost at the support is reduced. As a result, we can see that the order of anchor loss quality factor is often greater than other damping mechanisms in flexible beams. Here, the order of Q_{AL} is found to be 10^8–10^{12}. Further, we also computed

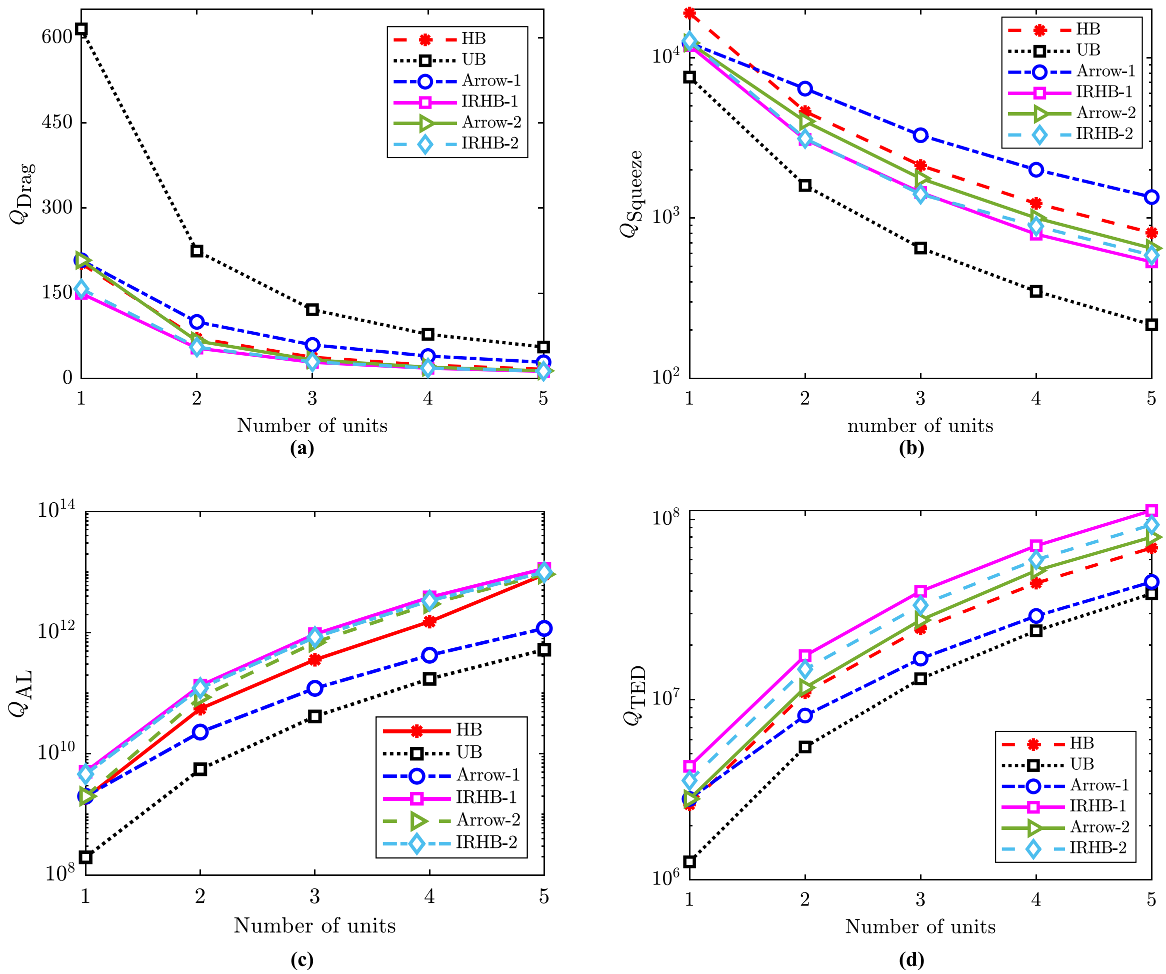

Fig. 10. Comparison of the numerically computed (a) Q_{Drag}, (b) Q_{Squeeze}, (c) Q_{AL}, and (d) Q_{TED} in Coventorware.

the thermoelastic damping (Q_{TED}), and the results have been shown in Fig. 10(d). It is observed that Q_{TED} is directly proportional to the length of the beams. As a result, Thermoelastic damping is directly related to the frequency of vibration. As the microbeams vibrate at higher frequencies, the impact of thermoelastic damping becomes more pronounced. and it is evident from Fig. 10(d) that the order of increasing Q_{TED} is as follows: UB < arrow-1 < HB < arrow-2 < IRHB-2 < IRHB-1. from the computed Q_{TED}, it is observed that as the flexibility of the beams increases, the temperature gradients across the cross-section of the beam decrease. As a result, there is reduced energy loss through thermoelastic damping, manifesting as a decrease in energy dissipation in the form of irreversible heat.

Hence, from the above discussion, it has been concluded that drag force damping is the dominant damping mechanism over others. Additionally, we calculated the numerical drag force damping quality factor for various shapes using the Ansys numerical tool, as detailed in the previous section. The outcomes of the numerical simulation are presented in Table 4. The results obtained from our analysis reveal that the drag force damping quality factor, denoted as Q_{Drag}, consistently decreases as the number of units increases from 1 to 5. This observed reduction in Q_{Drag} can be attributed to the increased surface area available for interacting with the oscillating beam and air particles. As the number of units grows, the expanded surface area facilitates a greater engagement between the oscillating structure and the surrounding air, leading to enhanced drag forces and, consequently, a decrease in the damping quality factor. This insight necessitates the importance of considering the number of units in the system when assessing the impact of drag force damping on oscillatory behavior. Among the various shapes considered, equivalent uniform beams (UB) exhibit a higher quality factor, primarily due to their inherent stiffness. As beams increase in stiffness, their natural frequency also increases. This increased frequency results in minimal interaction between the structure and the surrounding fluid, which leads to the increment in the Q_{Drag}. Nevertheless, hexagonal beams (HB) exhibit greater flexibility when compared to their equivalent uniform counterparts. The data presented in Table 4 illustrates that hexagonal beams have a higher drag force damping quality factor, denoted as Q_{Drag}, in comparison to arrow and irregular hexagons with similar stiffness characteristics. This difference in quality factor highlights the influence of structural geometry on the computation drag force damping, underscoring the unique advantages of hexagonal beams in reducing the damping effects.

4.5. Fabricated device

To extend the application of hexagon beams, we have fabricated two different configurations of a microdevice consisting of a square plate suspended by four double hexagonal microcantilever beams as shown in Fig. 11. In Structure 1, while the end of the second unit cell connects to the square membrane at the center, the truncated second unit cell connects to the square plate of Structure 2 at two equidistant points on the side of the relatively larger square membrane. Using the mechanical properties of silicon oxide, the numerical values of the first four frequencies and their flexible modes of structures 1 and 2 are

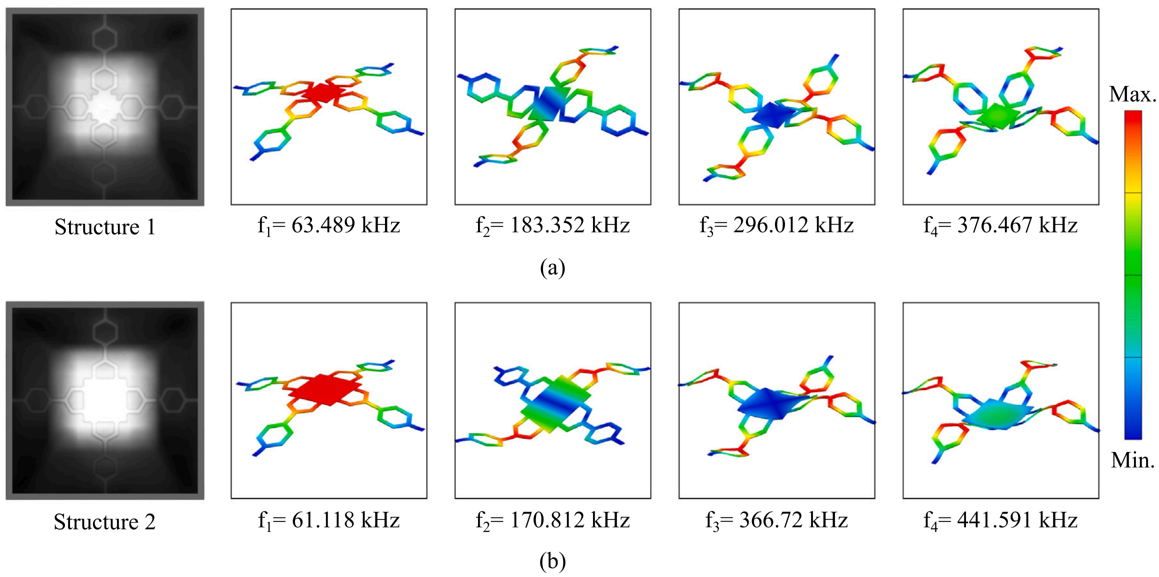

Fig. 11. (a) An optical image and numerical modeshapes and frequencies of structure 1; (b) An optical image and numerical modeshapes and frequencies of structure 2.

Table 4
Comparision of numerical Q_{Drag} for different beam shapes using Ansys software.

Configuration/Number of units	Q_{Drag}				
	1	2	3	4	5
HB	51.78	18.44	9.65	5.66	3.98
UB	97.12	43.85	26.33	17.27	12.64
Arrow-1	44.77	20.82	12.36	8.29	5.97
Arrow-2	44.77	16.44	8.63	5.31	3.72
IRHB-1	39.23	12.95	7.11	4.05	2.78
IRHB-2	40.96	14.08	7.32	4.48	3.08

shown in Fig. 11. The first mode exhibits out-of-plane motion of the middle membrane. The second mode shows the torsional motion of the membrane. In the third and fourth modes, the plate nearly remains static, while the beams show out-of-phase mode in mode 3 and in-phase mode in mode 4. The comparison of the modes of structures 1 and 2 shows that the first two modes are lowered while the next two higher modes are increased. To compare the results with experimental values, the experiments are performed at lower pressure of $p_1 = 4.8\times10^{-6}$ mbar. It is found that the experimental frequencies of the first transverse modes of both structures 1 and 2 are not observed, perhaps due to high damping. Therefore, the perforations with optimal size can be used to reduce damping. However, all the next three higher modes are observed experimentally and found to be in close agreement with the numerical results as mentioned in Table 5.

To see the influence of pressure on the structure, we perform experiments under two pressure conditions of $p_1 = 4.8 \times 10^{-6}$ mbar and $p_2 = 1.7 \times 10^{-2}$ mbar as mentioned in Table 6. A similar trend is observed for structures 1 and 2 in which the reduction in frequency of mode 2 is found to be more at lower pressure, p_1, than the higher pressure p_2. While frequency in mode 3 increases for structure 2 at lower pressure p_1, it increases at relatively higher pressure p_2. However, the frequencies of structure 2 increase at both pressure conditions. On measuring the frequencies of these structures at atmospheric pressure p_3, the first two modes are not observed, but the third mode frequencies for structures 1 and 2 are found as 260.586 kHz and 261.681 kHz, respectively. It concludes that the change in pressure from lower ($p_1 = 4.8 \times 10^{-6}$ mbar) to higher ($p_3 = 1013.25$ mbar) values reduces the third mode frequencies for structures 1 and 2 by 32.295 kHz and 135.404 kHz. It shows that Structure 2 is very sensitive to pressure. Further study is needed to understand the influence of pressure on the higher modes of such structures.

In the following section, we explore the potential applications of hexagonal beams and their equivalent uniform counterparts within the domain of MEMS accelerometers.

5. Application in MEMS accelerometer

In this section, we explore using various microbeams as suspension springs in out-of-plane MEMS accelerometers, also called z-axis accelerometers. These devices are specifically designed to measure acceleration in the vertical or out-of-plane direction. To initiate our discussion, we examine a MEMS accelerometer comprising a proof mass, sensing combs, and suspension springs, illustrated in Fig. 12.

The dimensions of the individual components of the device are detailed in Table 7. In this analysis, we employed the MEMS+ numerical tool to simulate the performance of the device. The structural material chosen for simulation was Si(100), and its material properties include a Young's Modulus (E) of 169 GPa, a Poisson's ratio (ν) of 0.28, and a density (ρ) of 2331 kg/m^3. Table 8 compares the stiffness, transverse mode frequency, and torsional mode frequency for the designed MEMS accelerometer with various suspension springs. These springs include the uniform beam (UB), hexagonal beam, irregular hexagonal beam-1 (IRHB-1), irregular hexagonal beam-2 (IRHB-2), and arrow microbeams. In our analysis, we have taken the width of each side of the hexagonal beam (w_s) to be 5 μm. To assess the impact of side width in the case of the uniform beam-2 (UB-2), we have also set the width at 5 μm for comparative purposes. From the Table 8, it is observed that for the same length and width, uniform beam (UB-1) has higher stiffness and natural frequency as compared to other structures. The hexagonal beam exhibits a substantial reduction in stiffness, transverse mode frequency, and torsional mode frequency by 89.17%, 66.4%, and 74.54%, respectively, when compared to UB-1. Hence, We considered a hexagonal-based MEMS accelerometer for further analysis.

To begin with, we performed the Modal analysis of the MEMS accelerometer with uniform and hexagonal beams as the suspension springs in MEMS plus the numerical tool. In Fig. 13(a) and (b) present a comparative analysis of the transverse and torsional mode frequencies of a MEMS accelerometer. The accelerometer is characterized by different beam widths, specifically $W_b = 35$ μm in Fig. 13(a), and $W_b = 155$ μm in Fig. 13(b). Examining these results allows for a clear understanding of how the transverse and torsional mode frequencies vary with the change in beam width. This comparison is necessary for understanding the impact of different beam widths on the dynamic

Table 5
A comparison between numerical frequencies and experimental frequencies at $p_1 = 4.8 \times 10^{-6}$ mbar for first four modes of structures 1 and 2.

Configuration	Mode 1 (f_1 in kHz)		Mode 2 (f_2 in kHz)		Mode 3 (f_3 in kHz)		Mode 4 (f_4 in kHz)	
	Num.	Exp.	Num.	Exp.	Num.	Exp.	Num.	Exp.
Structure 1	63.489	–	183.352	183.137	296.012	292.881	376.467	383.918
Structure 2	61.118	–	170.812	170.952	366.720	397.085	441.591	425.098

Table 6
Experimental values of frequencies for first four modes of structures 1 and 2 at pressure $p_1 = 4.8 \times 10^{-6}$ mbar and $p_2 = 1.7 \times 10^{-2}$ mbar.

Configuration	Mode 1 (f_1 in kHz)		Mode 2 (f_2 in kHz)		Mode 3 (f_3 in kHz)		Mode 4 (f_4 in kHz)	
	p_1	p_2	p_1	p_2	p_1	p_2	p_1	p_2
Structure 1	–	–	183.137	181.553	292.881	293.667	383.918	329.014
Structure 2	–	–	170.952	171.152	397.085	280.068	425.098	397.695

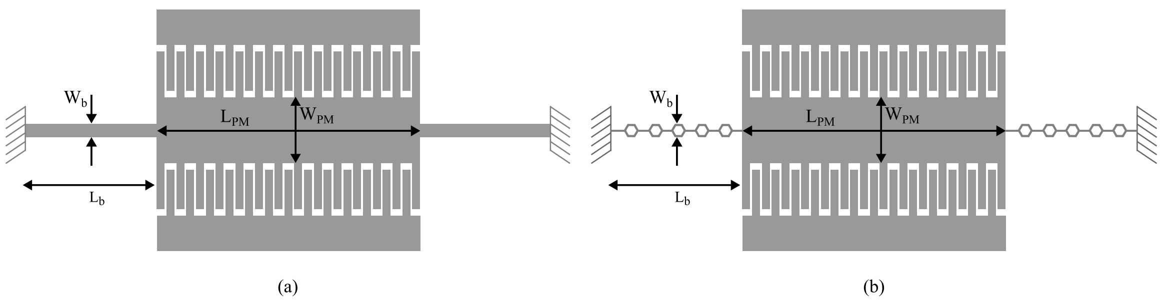

Fig. 12. Schematic of MEMS accelerometer with (a) uniform rectangular, and (b) hexagonal microbeams as the suspensions springs.

Table 7
Parameters of the MEMS accelerometer.

S.No	Parameter	Symbol	Value (μm)
1	Length of the proof mass	L_{PM}	600
2	Width of the proof mass	W_{PM}	155
3	Length of the suspension spring	L_b	300
4	Width of the suspension spring	W_b	35
5	Thickness of the proof mass	t_{PM}	10
6	Thickness of the suspension spring	t_b	10
7	Width of each side of the hexagonal spring	w_s	5
8	No. of comb pairs on single side	N_c	25
9	Overlap area	A_0	90 × 10 μm^2

behavior of the MEMS accelerometer, providing insights into the system response under transverse and torsional forces. The findings indicate that, owing to enhanced flexibility, the MEMS accelerometer based on the hexagonal beam has a lower natural frequency in both transverse and torsional modes than its uniform beam-based counterpart with the same width. Therefore, based on our earlier comprehension, we investigated the MEMS accelerometer by employing hexagonal beams with different widths and thicknesses, as illustrated in Fig. 14. In Fig. 14(a), (b), and (c), the MEMS accelerometer is depicted with hexagonal microbeams having widths of 65 μm, 95 μm, and 155 μm, respectively. The corresponding frequencies of the transverse and torsional modes, with varying thickness, are presented in Fig. 14(d). From Fig. 14(d), it has been observed that the natural frequencies of both transverse and torsional modes increase with an increase in the width and thickness of the suspension springs. This is attributed to the increased stiffness of the hexagonal suspension springs resulting from changes in width and thickness. In other words, as the width and thickness of the suspension springs increase, so does their stiffness, consequently leading to an increase in the natural frequencies of both transverse and torsional modes.

Additionally, we conducted a comprehensive analysis using MEMS plus software to assess both static and electromechanical aspects of the designed MEMS accelerometer. This evaluation was performed under the influence of an input acceleration of ±30 g in the transverse direction, where "g" represents the acceleration due to gravity (9.81 m/s^2). The objective was to examine the behavior of the device and response to varying levels of acceleration, providing insights into its performance characteristics under different operational conditions. We have used the area-changing capacitive-based method for the sensing, which is shown in the Fig. 12(b). When the device experiences acceleration in the Z-axis direction, the proof mass tends to move relative to the fixed electrodes. Then it forms a relative area change in capacitance due to inertial forces applied by external input acceleration.

The investigation involved a detailed hexagonal beam (HB) based MEMS accelerometer analysis, focusing on two critical performance parameters: transverse displacement and capacitance. In the first aspect, we explored the relationship between transverse displacement (measured in nanometers) and input acceleration (expressed in gravitational units, g) for varying widths of the hexagonal beam, as shown in Fig. 15(a). The results show that the proof mass displacement decreases as the width of the bean increases from 35 μm to 155 μm due to an increase in the transverse stiffness of the hexagonal beams. Simultaneously, we did the electromechanical analysis for the change in capacitance (measured in femtofarads, fF) with an input acceleration (in g) on the hexagonal beam-based MEMS accelerometer, specifically focusing on the influence of different beam widths as shown in Fig. 15(b). The results we obtained reveal a static capacitance of 287.5676 fF. Additionally, it was observed that the change in capacitance decreases as the acceleration increases from zero to ±30 g. This decrease is attributed to the reduction in the overlap area as the input acceleration increases in the transverse direction.

Further, we also explored the gap-changing differential capacitive sensing mechanism in the MEMS accelerometer. The differential capacitance (ΔC) in a MEMS accelerometer is influenced by the gap (d) between the two parallel plates, which are typically formed by a movable proof mass and stationary electrodes as shown in Fig. 16(a). As the proof mass experiences acceleration in the transverse direction, it moves relative to the stationary electrodes (top and bottom electrodes), changing the gap between the proof mass and electrodes. This change

Table 8
Transverse and torsional mode frequency corresponds to the MEMS accelerometer using different beams as suspension springs.

S.No	Beam	Stiffness (N/m)	Transverse mode frequency (kHZ)	Torsional mode frequency (kHZ)
1	UB-1	438.14	59.88	76.28
2	Hex	47.45	20.11	19.42
3	IRHB-1	38.52	18.06	15.32
4	IRHB-2	41.39	18.75	16.66
5	Arrow	52.32	20.87	17.14
6	UB-2	62.59	23.22	13.53

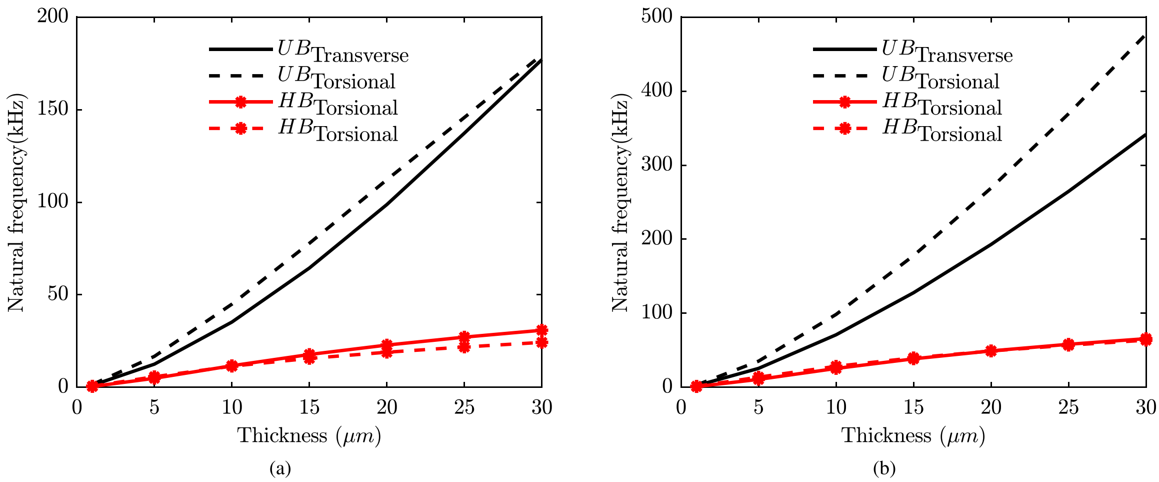

Fig. 13. Comparision of (a) Transverse and torsional frequency of uniform beam (UB) and hexagonal beam (HB) for the length (L_b = 300 μm) and width (W_b = 35 μm), (b) Transverse and torsional frequency of uniform beam (UB) and hexagonal beam (HB) for the length (L_b = 300 μm) and width (W_b = 155 μm).

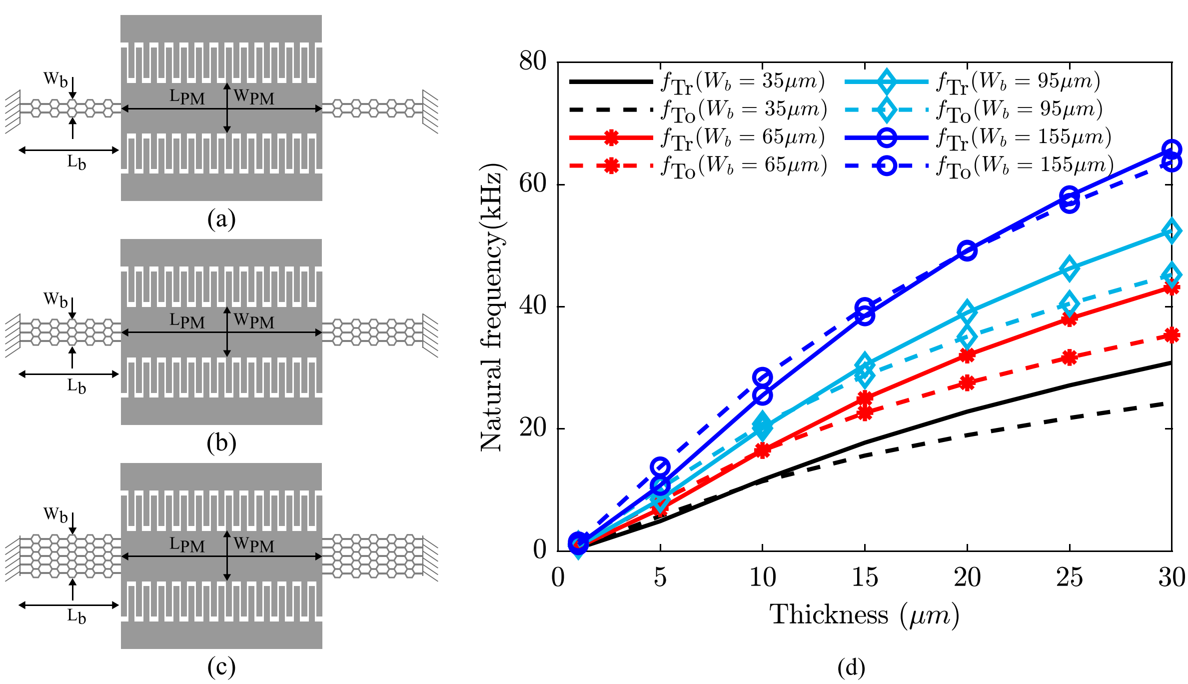

Fig. 14. The MEMS accelerometer with hexagonal beams as the suspensions springs with width of the beams (a) W_b = 65 μm, (b) W_b = 95 μm, (c) $W_b = W_{PM}$ = 65 μm, and (d) the corresponding transverse (f_{Tr}) and torsional (f_{To}) mode frequency with varying thickness of the beam.

in gap alters the capacitance, which is then measured to determine the input acceleration applied over the MEMS accelerometer. When the proof mass undergoes transverse motion induced by inertial forces, an important phenomenon called squeeze film damping comes into play. In this scenario, the air particles within the gap between the electrode plates and the proof mass become compressed. This compression of the gas introduces a damping mechanism, leading to the dissipation of energy. Specifically, as the gap between the moving proof mass and the stationary electrodes changes, the air particles experience compression, generating a damping force that acts counter to the motion. This process, known as squeeze film damping, plays a crucial role in dissipating energy within the MEMS accelerometer during transverse motion. In Fig. 16(b), the stiffness (K) in N/m and damping ratio (ζ) of the MEMS accelerometer is illustrated for different widths of the

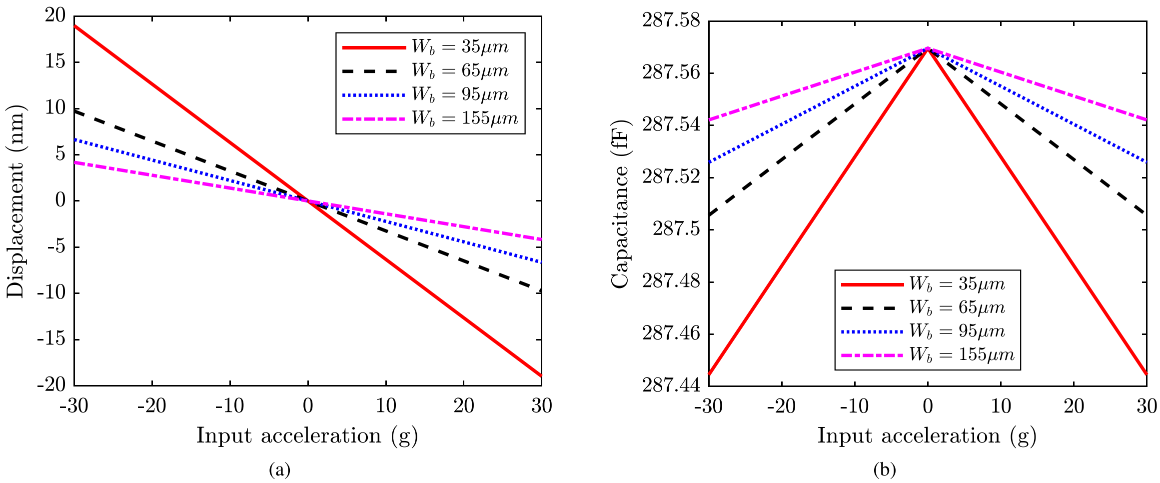

Fig. 15. Comparision of (a) Transverse displacement (nm) Vs input acceleration (g) of the hexagonal beam (HB) based MEMS accelerometer for varying width of the beam, (b) Capacitance (fF) Vs input acceleration (g) of the hexagonal beam (HB) based MEMS accelerometer for varying width of the beam.

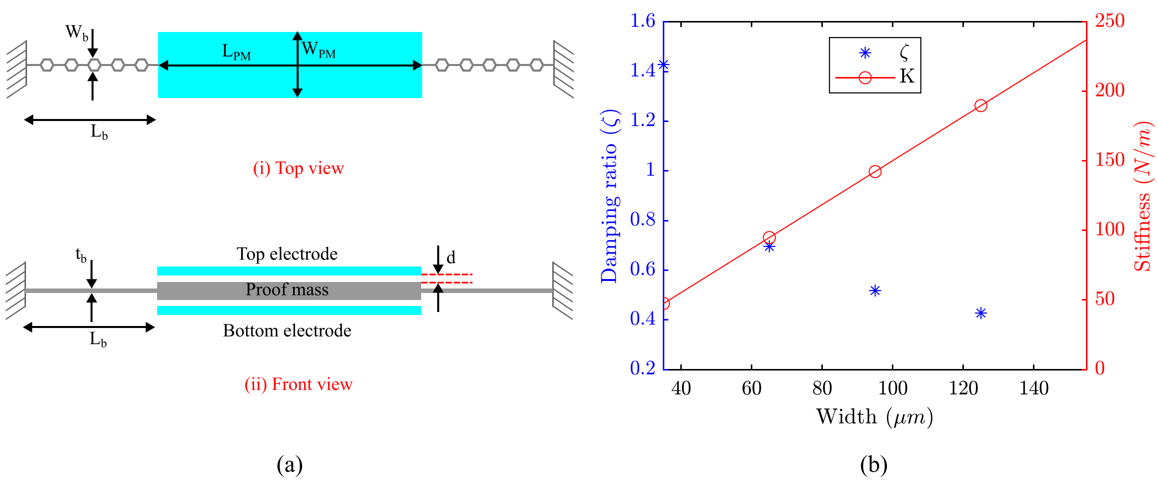

Fig. 16. The MEMS accelerometer with hexagonal beams as the suspensions spring with width of the beams, W_b = 35 μm, W_b = 65 μm, W_b = 95 μm, W_b = 125 μm, $W_b = W_{PM}$ = 155 μm, (a) Schematic of top and front view of MEMS accelerometer, and (b) Corresponding damping ratio (ζ), and stiffness (K) for varying widths.

hexagonal suspension beams. As the width of the beams ranges from 35 μm to 155 μm, the transverse stiffness of the device increases and decreases in the associated damping ratio. Therefore, there has been a trade-off between the flexibility and quality factor ($Q = \frac{1}{2\zeta}$) of MEMS accelerometers when they operate in an ambient environment.

Following that, we conducted static and electromechanical analyses on the out-of-plane MEMS accelerometer, as depicted in Fig. 17(a), considering an input acceleration range of ±30 g. Fig. 17(a) and (b) present the proof mass displacement and the corresponding differential capacitance ($\Delta C = C_2 - C_1$) of the accelerometer, respectively. Here, C_1 represents the capacitance formed between the proof mass and the bottom electrode, while C_2 corresponds to the capacitance formed between the proof mass and the top electrode. As depicted in Fig. 17(a), it can be observed that the displacement of the proof mass diminishes with the widening of the suspension springs, ranging from 35 μm to 155 μm for an input acceleration of ±30 g, this is due to the increase in the transverse stiffness of the suspension springs. Similarly, Fig. 17(b) shows corresponding differential capacitance change (ΔC) in fF; as the suspension springs have become wider, the value of ΔC reduces due to an increase in the transverse stiffness.

In conclusion, the extensive analysis and discussion presented above lead to the finding that hexagonal microbeams hold great promise for application as suspension springs in MEMS accelerometers. The comparative assessment with equivalent rectangular uniform microbeams underscores the superior flexibility and vibrational characteristics inherent in the hexagonal microbeam design. This study suggests that implementing hexagonal microbeams as suspension springs can enhance MEMS accelerometer overall performance and functionality, making them more suitable for accurate sensing and responding to dynamic forces. The unique geometric attributes of hexagonal microbeams contribute to improved flexibility and vibrational behavior, offering valuable insights for advancements in MEMS accelerometer technology.

6. Conclusions

In this work, we have analyzed a new design of microcantilever beams with hexagonal shapes. The proposed microstructures are fabricated using a silicon-wet bulk micromachining process. The new designs are analyzed for natural frequency, maximum deflection, and Q-factor in the first transverse mode of vibration and in comparison with the equivalent uniform rectangular microcantilever beams. Various damping mechanisms like drag force damping, squeeze film damping, anchor loss damping, and thermoelastic damping are considered for the analysis of the Q-factor of the microbeams. Experimental first mode natural frequency has been measured for the proposed designs and compared with the numerical first natural frequency, which

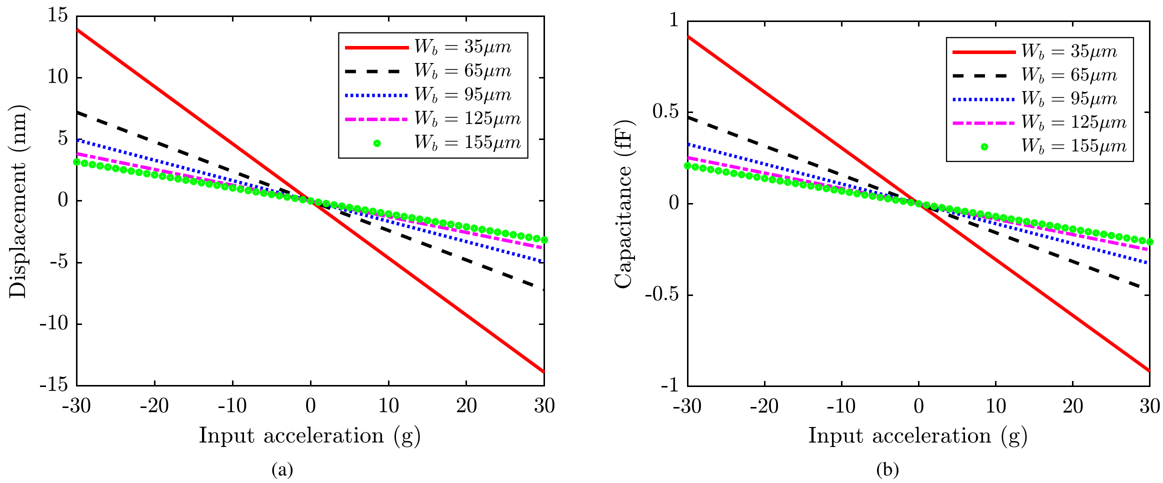

Fig. 17. Comparision of (a) Transverse displacement (nm) Vs input acceleration (g) of the hexagonal beam (HB) based MEMS accelerometer for varying width of the beam, (b) Capacitance (fF) Vs input acceleration (g) of the hexagonal beam (HB) based MEMS accelerometer for varying width of the beam,.

has shown good agreement. As the microbeam configuration changes from a single hexagon to a quintuple hexagon, the natural frequency is reduced due to the reduction in effective stiffness or increase in the effective mass of the microbeams. It is also shown that the natural frequency of the hexagonal microcantilever beams is lower than the equivalent uniform rectangular beams due to a reduction in stiffness. Further, we numerically computed the maximum static deflection of hexagonal and uniform beams for the concentrated load of 10 nN at the free end. As the beam length changes from single to quintuple, the maximum deflection (in nm) of the beams increases due to a reduction in stiffness or an increase in effective mass. The maximum static deflection of the hexagonal beams is higher than the uniform beams due to reduced stiffness. Further, we analyzed the quality factor (Q-factor) of the new designs for different damping mechanisms. Among all the damping mechanisms, drag force damping is more dominant with the order of magnitude as 10^1–10^2. In addition, we validated our experimentally measured quality factor with the semi-analytical boundary element method and numerical Ansys technique proposed in the existing literature, which has shown good agreement with the same order of magnitude. Hence, the newly proposed hexagonal microcantilever beams can be employed in various MEMS applications due to their flexibility and damping characteristics.

CRediT authorship contribution statement

Sai Kishore Jujjuvarapu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Lalsingh Devsoth:** Writing – review & editing, Visualization, Data curation. **Ashok Akarapu:** Writing – review & editing, Visualization, Methodology, Data curation. **Prem Pal:** Writing – review & editing, Visualization. **Ashok Kumar Pandey:** Writing – review & editing, Visualization, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The first author would like to acknowledge the fellowship provided by the Ministry of Education, Government of India, and partial support from the project sponsored by DRDO, New Delhi, India.

References

[1] F. Ulucan-Karnak, C.İ. Kuru, S. Akgol, Sensor commercialization and global market, Adv. Sens. Technol. (2023) 879–915.
[2] Tilli, Paulasto-Kröckel, Petzold, Theuss, Motooka, Lindroos, Handbook of Silicon Based MEMS Materials and Technologies, Elsevier, 2020.
[3] Moulin, O'shea, M.E. Welland, Micro cantilever-based biosensors, Ultramicroscopy 82 (1-4) (2000) 23–31.
[4] Alleja, Montserrat, Highly sensitive polymer-based cantilever-sensors for DNA detection, Ultramicroscopy 105 (1-4) (2005) 215–222.
[5] A. Mana, N. Jalili, Towards non-linear modeling of molecular interactions arising from adsorbed biological species on the micro cantilever surface, Int. J. Non-Linear Mech. 42 (4) (2007) 588–595.
[6] A. Mar, L.M. Lechuga, Micro cantilever-based platforms as biosensing tools, Analyst 135 (5) (2010) 827–836.
[7] G. P., V. Anitha, S.A. Mastani, Micro cantilever based biosensor for disease detection applications, J. Med. Bioeng. 34 (2015).
[8] T. Yuan, A low spring constant piezoresistive micro cantilever for biological reagent detection, Micromachines 11 (11) (2020) 1001.
[9] T. Bali, M. Jadhav, Design and exploration of micro cantilever biosensor for detection of tuberculosis, ECS Trans. 107 (1) (2022) 627.
[10] A.M. Zahid, C. Cho, Design and analysis of a high sensitive micro cantilever biosensor for biomedical applications, in: International Conference on BioMedical Engineering and Informatics, Vol. 2, IEEE, 2008.
[11] W. Qu, Micro cantilever array pressure measurement system for biomedical instrumentation, IEEE Sens. J. 10 (2) (2009) 321–330.
[12] K. Farbod, C.W. de Silva, Recent advances in MEMS sensor technology–biomedical applications, IEEE Instrum. Meas. Mag. 15 (1) (2012) 8–14.
[13] S.K. Kumar, T. Muthumanickam, MEMS technology for early stage diesis detection using micro-cantilever structure in support of biomedical applications, ECS Trans. 107 (1) (2022) 1125.
[14] P. Vinayak, Precise design of micro-cantilever sensor for biomedical application, Sens. Lett. 18 (12) (2020) 900–904.
[15] G. Catherine, Silicon micro cantilever sensors to detect the reversible conformational change of a molecular switch, Sensors 20 (3) (2020) 854.
[16] L.G. Sai, Design and performance analysis of a microbridge and micro cantilever-based MEMS pressure sensor for glucose monitoring, IEEE Sens. J. 23 (5) (2023) 4589–4596.
[17] D.R. Baselt, Design and performance of a micro cantilever-based hydrogen sensor, Sensors Actuators B 88 (2) (2003) 120–131.
[18] P. Wu, N. Li, Micro-cantilever array and its application in gas sensor, in: International Conference on Microwave and Millimeter Wave Technology, Vol. 3, IEEE, 2008, pp. 1547–1550.
[19] T. Sébastien, Modeling and performance of uncoated micro cantilever-based chemical sensors, Sensors Actuators B 143 (2) (2010) 555–560.
[20] D. Ying, Characterization of the gas sensors based on polymer-coated resonant micro cantilevers for the detection of volatile organic compounds, Anal. Chim. Acta 671 (1-2) (2010) 85–91.

[21] B. Mohand-Tayeb, Geometry optimization of uncoated silicon micro cantilever-based gas density sensors, Sensors Actuators B 208 (2015) 600–607.
[22] B. Yuyang, Detection of volatile-organic-compounds (VOCs) in solution using cantilever-based gas sensors, Talanta 182 (2018) 148–155.
[23] X. Dongcheng, A low power cantilever-based metal oxide semiconductor gas sensor, IEEE Electron Device Lett. 40 (7) (2019) 1178–1181.
[24] D. Hélène, I. Dufour, Resonant micro cantilever devices for gas sensing, in: Advanced Nanomaterials for Inexpensive Gas Microsensors, Elsevier, 2020, pp. 161–188.
[25] W. Jingjing, Micro cantilever sensors for biochemical detection, J. Semicond. 44 (2) (2023) 023105.
[26] R. Daniel, Measurement of the mass and rigidity of adsorbates on a micro cantilever sensor, Sensors 7 (9) (2007) 1834–1845.
[27] N. Margarita, Sensitivity improvement of a micro cantilever based mass sensor, Microelectron. Eng. 86 (4-6) (2009) 1187–1189.
[28] N. Margarita, A high sensitivity silicon micro cantilever based mass sensor, Sensors (2008) 1127–1130.
[29] R.K. Godara, A novel capacitive mass sensor using an open-loop controlled micro cantilever, Microsyst. Technol. 26 (2020) 2977–2987.
[30] L. Kai, et al., 4D printed zero Poisson's ratio metamaterial with switching function of mechanical and vibration isolation performance, Mater. Des. 196 (2020) 109153.
[31] A.R. Mohammed, H. Abdulhadi, A. Mian, Advances in mechanical metamaterials for vibration isolation: A review, Adv. Mech. Eng. 14 (3) (2022) 16878132221082872.
[32] S. Ahmad, et al., 4D printed shape memory sandwich structures: experimental analysis and numerical modeling, Smart Mater. Struct. 31 (5) (2022) 055014.
[33] Q. Qing, I. Dayyani, P. Webb, Structural Mechanics of cylindrical fish-cell zero Poisson's ratio metamaterials, Compos. Struct. 289 (2022) 115455.
[34] Shimada, V. Lich, N. Wang, Kitamura, Hierarchical ferroelectric and ferrotoroidic polarizations coexistent in nano-metamaterials, Sci. Rep. 5 (2015).
[35] Chen, Jia, Wang, Hierarchical honeycomb lattice metamaterials with improved thermal resistance and mechanical properties, Compos. Struct. 152 (2016) 395–402.
[36] Usta, Scarpa, Türkmen, Edgewise compression of novel hexagonal hierarchical and asymmetric unit cells honeycomb metamaterials, Mater. Today Commun. 24 (2020).
[37] Xu, Zhao, Wang, Duan, Lei, Fang, Mechanical performance of bio-inspired hierarchical honeycomb metamaterials, Int. J. Solids Struct. (2022) 254–255.
[38] C. Sparavigna, Paper-based metamaterials: Honeycomb and auxetic structures, Int. J. Sci. 11 (2014) 22–25.
[39] Ebrahimi-Nejad, Kheybari, Honeycomb locally resonant absorbing acoustic metamaterials with stop band behavior, Mater. Res. Express 10 (2018).
[40] Jiang, et al., Electromagnetic wave absorption and compressive behavior of a three-dimensional metamaterial absorber based on 3D printed honeycomb, Sci. Rep. 8 (2018).
[41] Li, Zhang, Xie, A lightweight multilayer honeycomb membrane-type acoustic metamaterial, Appl. Acoust. 168 (2020).
[42] Namvar, Zolfagharian, Vakili-Tahami, Bodaghi, Reversible energy absorption of elasto-plastic auxetic, hexagonal, and AuxHex structures fabricated by FDM 4D printing, Smart Mater. Struct. 31 (2022).
[43] Liu, Zhang, Chang, Lyu, Zhao, Qiu, Programmable mechanical metamaterials: basic concepts, types, construction strategies—a review, Front. Mater. 11 (2024).
[44] G. Aparna, A. Ashok, C.S. Sharma, P. Pal, A.K. Pandey, Frequency analysis of hexagonal microbeam with 2D nanofiber mat, Mater. Res. Express 6 (8) (2019) 085631.
[45] Kurmendra, R. Kumar, A review on RF micro-electro-mechanical-systems (MEMS) switch for radio frequency applications, Microsyst. Technol. 27 (7) (2021) 2525–2542.
[46] A. Ashutosh, S. Pal, S. Kundu, Multi-perforated energy-efficient piezoelectric energy harvester using improved stress distribution, IETE J. Res. 69 (6) (2023) 3723–3738.
[47] P.-C. Angel, et al., Optimization of Q-factor of AFM cantilevers using genetic algorithms, Ultramicroscopy 115 (2012) 61–67.
[48] M.A. Hamed, N.A. Mohamed, M.A. Eltaher, Stability buckling and bending of nanobeams including cutouts, Eng. Comput. 38 (1) (2020) 209–230.
[49] D.P. Giorgio, T. Veijola, A. Somà, Modelling and validation of air damping in perforated gold and silicon MEMS plates, J. Micromech. Microeng. 20 (1) (2009) 209–230.
[50] B. Hicham, Analytical modeling for the determination of nonlocal resonance frequencies of perforated nanobeams subjected to temperature-induced loads, J. Micromech. Microeng. 75 (2016) 163–168.
[51] L. Pu, R. Hu, A model for squeeze-film damping of perforated MEMS devices in the free molecular regime, J. Micromech. Microeng. 21 (2) (2011).
[52] L. Pu, Y. Fang, An analytical model for squeeze-film damping of perforated torsional microplates resonators, Sensors 15 (4) (2015) 7388–7411.
[53] A.M. Zahid, C. Cho, Deflection, frequency, and stress characteristics of rectangular, triangular, and step profile micro cantilevers for biosensors, Sensors 9 (8) (2009) 6046–6057.
[54] L. Yan, Geometry and profile modification of micro cantilevers for sensitivity enhancement in sensing applications, Sensors Mater. 29 (1) (2017).
[55] P.D. Kishore, J. Singh, P.K. Kankar, Variable width based stepped MEMS cantilevers for micro or pico level biosensing and effective switching, J. Mech. Sci. Technol. 29 (2015) 4823–4832.
[56] H. Rouhollah, M. Hamedi, Resonant frequency of bimorph triangular V-shaped piezoelectric cantilever energy harvester, J. Comput. Appl. Res. Mech. Eng. (JCARME) 6 (1) (2016) 65–73.
[57] S.S. Singh, P. Pal, A.K. Pandey, Mass sensitivity of non-uniform micro cantilever beam, ASME J. Vib. Acoust. 128 (6) (2016) 064502.
[58] S.S. Singh, P. Pal, A.K. Pandey, Pull-in analysis of non-uniform micro cantilever beams under large deflection, J. Appl. Phys. 118 (20) (2015) 204303.
[59] A. Akarapu, et al., Design and analysis of micro cantilever beams based on arrow shape, Microsyst. Technol. 25 (2019) 4379–4390.
[60] A. Akarapu, et al., An analysis of stepped trapezoidal-shaped micro cantilever beams for MEMS-based devices, J. Micromech. Microeng. 28 (7) (2018) 075009.
[61] D. Lalsingh, A.K. Pandey, Hydrodynamic forces in non-uniform cantilever beam resonator, Int. J. Mech. Sci. 244 (2023) 108078.
[62] W. Famei, M. Zou, Changrui, et al., Three-dimensional printed microcantilever with mechanical metamaterial for fiber-optic microforce sensing, APL Photonics 8 (9) (2023).
[63] H. Hiroshi, K. Itao, S. Kuroda, Damping characteristics of beam-shaped micro-oscillators, Sensors Actuators A 49 (1-2) (1995) 87–95.
[64] J.A. Judge, Attachment loss of micromechanical and nanomechanical resonators in the limits of thick and thin support structures, J. Appl. Phys. 101 (1) (2007).
[65] F.R. Blom, Dependence of the quality factor of micromachined silicon beam resonators on pressure and geometry, J. Vac. Sci. Technol. B 10 (1) (1992) 19–26.
[66] G.W. S., H.H. Richardson, S. Yamanami, A study of fluid squeeze-film damping, J. Fluids Eng. (1966) 451–456.
[67] R. Lifshitz, M.L. Roukes, Thermoelastic damping in micro-and nanomechanical systems, Phys. Rev. B 61 (8) (2000) 5600.
[68] N.C. Cuong, W.L. Li, Effects of surface roughness and gas rarefaction on the quality factor of micro-beam resonators, Microsyst. Technol. 23 (2017) 3489–3504.
[69] N.C. Cuong, W.L. Li, Influences of temperature on the quality factors of micro-beam resonators in gas rarefaction, Sensors Actuators A 261 (2017) 151–165.

Jujjuvararapu Sai Kishore received a B.Tech degree in Mechanical Engineering from Rajiv Gandhi University of Knowledge Technologies, IIIT RK Valley, India, in 2015. He is currently pursuing a Ph.D. degree at the Indian Institute of Technology Hyderabad, India. His research interests include the design and fabrication of microelectromechanical systems.

Lalsingh Devsoth received the B.E. degree from Osmania University, Hyderabad, India, in 2012, MTech. degree in design engineering in Applied Mechanics Department from IIT Delhi, India in 2016, and the Ph.D. degree from Department of Mechanical and Aerospace Engineering, IIT Hyderabad, Hyderabad, India, in 2023. His research interests include microfluidics, MEMS and fluid structure interactions, AFM, biosensors and energy harvesting devices.

Akarapu Ashok received the B.Sc. degree from Osmania University, Hyderabad, India, in 2003; the M.Sc. degree in Physics from Osmania University, Hyderabad, India, in 2007; the M.Tech. degree from department of Metallurgical & Materials Engineering, NIT Rourkela, Rourkela, India, in 2011; and the Ph.D. degree from department of Physics, IIT Hyderabad, Hyderabad, India, in 2015. He worked as a Post-doctoral fellow in the department of Mechanical & Aerospace Engineering, IIT Hyderabad, Hyderabad, India. Currently, he is working as assistant professor in the department of physics at Mallareddy university, Hyderabad, India. His research interests include thin films, electrode position, anodic oxidation, wet bulk micromachining for MEMS and solar cells, micro mechanical beams for MEMS.

Prem Pal did his M.Tech. in Solid State Material (SSM) in Dec. 1999 and Ph.D. in the area of MEMS in Dec. 2004, both from Indian Institute of Technology Delhi (IIT Delhi), India. The Foundation for Innovation and Technology Transfer (FITT), IIT Delhi awarded his thesis for "Best Industry Relevant Ph.D. Project - 2005". From July 2005 to June 2006, he worked as Postdoctoral Researcher at Yonsei Microsystems Laboratory, School of Mechanical Engineering, Yonsei University Seoul, South Korea. He worked in Sato laboratory, Department of Micro-Nano Systems Engineering, Nagoya University, Nagoya, Japan: from July 2006 to March 2008 as COE Scientist and from March 2008 to March 2010 as JSPS Fellow. He joined the Department of Physics, Indian Institute of Technology Hyderabad in April 2010 as Assistant Professor. Presently, he is working as Professor. He received the Faculty Teaching Excellence Award-2024 from IIT Hyderabad. He has published more than 80 papers in peer reviewed international journals of high repute and many papers in international conferences. In addition, he has published one book entitled "Silicon Wet Bulk Micromachining for MEMS". His

research interests include MEMS technology, MEMS based sensors, Silicon and Glass Micromachining, Silicon Surface Texturing, Thin films for MEMS.

Ashok Kumar Pandey is currently a Professor and the Head of the Mechanical and Aerospace Engineering at Indian Institute of Technology Hyderabad. Before joining the Institute, he completed his master' and Ph.D. from Indian Institute of Science, Bangalore, and postdoctoral research from Israel Institute of Technology, Technion, Haifa. During his Ph.D., he spent six months in the Mechanical Engineering Department at National University of Singapore towards development of characterization setup for MEMS torsional mirrors under different ambient conditions. The related published work was awarded with Hetenyi Award 2010 by the Society of Experimental Mechanics, USA. He was the convener for nanoscience and nanotechnology for IITH-JICA friendship project since its inception till 2019 and also an officiating Dean, Academics at Indian Institute of Technology Bhilai until 2021. He has initiated and coordinated two major interdisciplinary master's program in Integrated Sensors System and Electrical Vehicle Technology at IIT Hyderabad. He has conducted several workshops in the area of MEMS and vehicle dynamics, and organized an international conference under the aegis of IFToMM. He introduced flip teaching in some of his courses for which he received the best teaching award by IIT Hyderabad in 2012. His primary area includes linear and nonlinear vibration, multiphysics modeling of different phenomena in MEMS resonant devices, vehicle and tire dynamics. He has published over 66 journals and over 126 talks in conferences and workshops. He has coedited a book on the proceedings of 6th International conference on Microactuator, Microsensor and Micromechanisms.